

Kingdom of Cambodia
Nation Religion King



EMR Clinical Data Integration Specification

V 1.1



National Payment Certification Agency
Prepared by Department of Data Management and Information Technology

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1 OVERVIEW

1.1 PURPOSE

This document defines the technical specification for submitting clinical visit data from **external EMR systems** to the **Digital Payment Certification System**. It ensures standardized, secure, and interoperable data exchange.

1.2 SCOPE

This specification covers:

- **Authentication and Security:** Secure communication protocols and authentication methods
- **Data Submission:** Complete the visit data submission process
- **Data Coverage:** Clinical, administrative, and billing-related information
- **Response Handling:** Acknowledgement (ACK) and processing behavior
- **Error Management:** Validation rules and error handling procedures

1.3 INTENDED AUDIENCE

- **EMR Vendors:** External system developers and implementers
- **Integration Developers:** Backend and API integration specialists
- **Technical Architects:** System designers and solution architects
- **Quality Assurance Teams:** Testing and interoperability validators

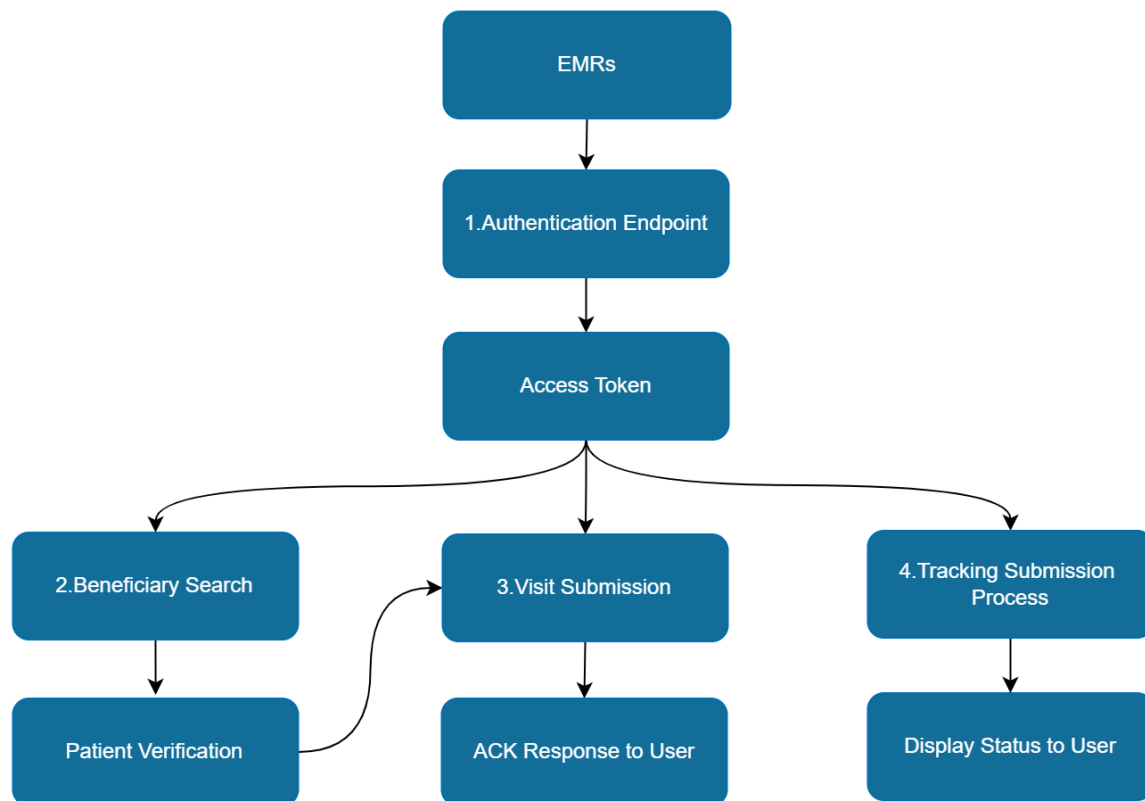
1.4 IMPLEMENTATION STANDARDS

- **API Protocol:** RESTful JSON over HTTPS
- **Diagnostic Coding:** ICD-10 (Category/Subcategory codes)
- **NPCA Data Dictionary**
- **Date/Time Format:** ISO-8601 with Cambodia time zone (+07:00)
 - *Example:* 2025-11-26T11:32:00.000+07:00
- **Demographic Standards:** National Committee for Demography and Development (NCDD) standards
 - *Reference data:* <https://data.mef.gov.kh/>

2 INTEGRATION ARCHITECTURE & DATA FLOW

2.1 ENDPOINT SEQUENCE & DEPENDENCIES

The integration involves three primary API endpoints that work together to manage patient visits:



Endpoint 1: Authentication

- **Purpose:** Obtain a secure OAuth 2.0 access token (Bearer token) for authenticating all subsequent API calls.
- **When to Use:** Before any API call; upon token expiration (via 401 Unauthorized response); or as part of a scheduled refresh cycle.
- **Frequency:** Initial setup and periodic refresh (tokens are typically valid for 2 hours).
- **Dependencies:** None. This is the entry point.

Endpoint 2: Beneficiary Search

- **Purpose:** Verify if the card exists, has a benefit program, and has basic demographic information.
- **When to Use:** **Strongly recommended** before visit submission to ensure data linkage and avoid errors.
- **Frequency:** Once per patient/visit verification.
- **Dependencies:** Requires a valid **access token** from Endpoint 1.
- **Output:** Critical Eligibility card, SPID used in the Visit Submission payload.

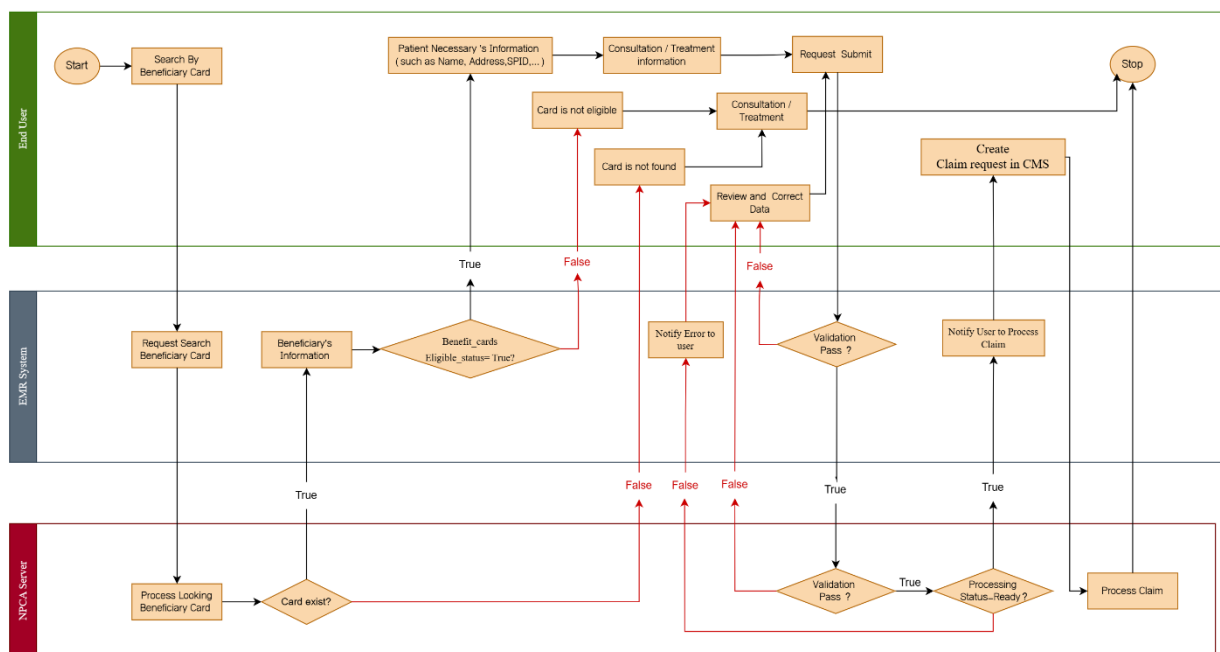
Endpoint 3: Visit Submission

- **Purpose:** Submit complete or partial clinical visit data for processing and storage.
- **When to Use:** After successful patient verification (Endpoint 2) and with a valid access token (Endpoint 1).
- **Frequency:** Once per patient visit event (create/update).
- **Dependencies:** Requires a valid **access token** (Endpoint 1) and a SPID (obtained via Endpoint 2).
- **Response:** Immediate HTTP 200 Accepted with a **status**, **visit_ulid** and **visit_code**.

Endpoint 4: Tracking Processing Result

- **Purpose:** Retrieve the latest processing status and validation result of a previously submitted visit, including whether it has been accepted, rejected, is still processing, or requires user action.
- **When to Use:** After a visit has been submitted to the Integration Platform (Endpoint 3) and with a valid access token (Endpoint 1). The EMR needs to check the current processing status or obtain the final processing result. This is especially useful when processing is asynchronous or when the initial submission response only acknowledges receipt.
- **Frequency:** On demand or periodically until processing is completed. Recommended polling interval: every 8 hours (or according to system guidelines). Stop polling once the status becomes **Completed**, **Rejected**, or **Failed**.
- **Dependencies:** Authorized API credentials and Visit Submission
- **Response:** Returns the **status** (**Processing**, **Pending**, **Failed**, **Ready**), processed timestamps, Validation or processing failure reason, processing status, and any identifiers such as Visit Code.

2.2 HOW EMR INTEGRATION WORKS



Flow of EMR Integration

Step 1: Beneficiary Lookup

This is where the process starts. User swipes or enters the patient's card number into the system.

- **Action:** User swipes or manually enters the patient's benefit card number.
- **Outcomes:**
 1. If the card is **found**, the system retrieves the **Patient's Details** (Home, Address, SPID, etc.). The "Patient's Information" is prepared, and the process moves to step 2.
 2. If the **Card is not found** in the database, the patient would **not be eligible to receive treatment under the benefit**.

Step 2: Eligibility & Card Validity Check

- **Action:** System verifies eligibility status linked to the patient's record.
- **Outcomes:**
 1. **Eligible:** Patient information is confirmed and prepared. The patient may proceed to receive covered treatment.
 2. **Not Eligible:** Benefit does **not apply**; treatment will not be covered.

Step 3: Benefit Request Submission

1. Submission Initiation

- OPD/IPD: Submit after treatment completion
- IPD-HEF-Poor: Submit after admission (pre-authorization/coverage guarantee)

2. EMR Portal Validation

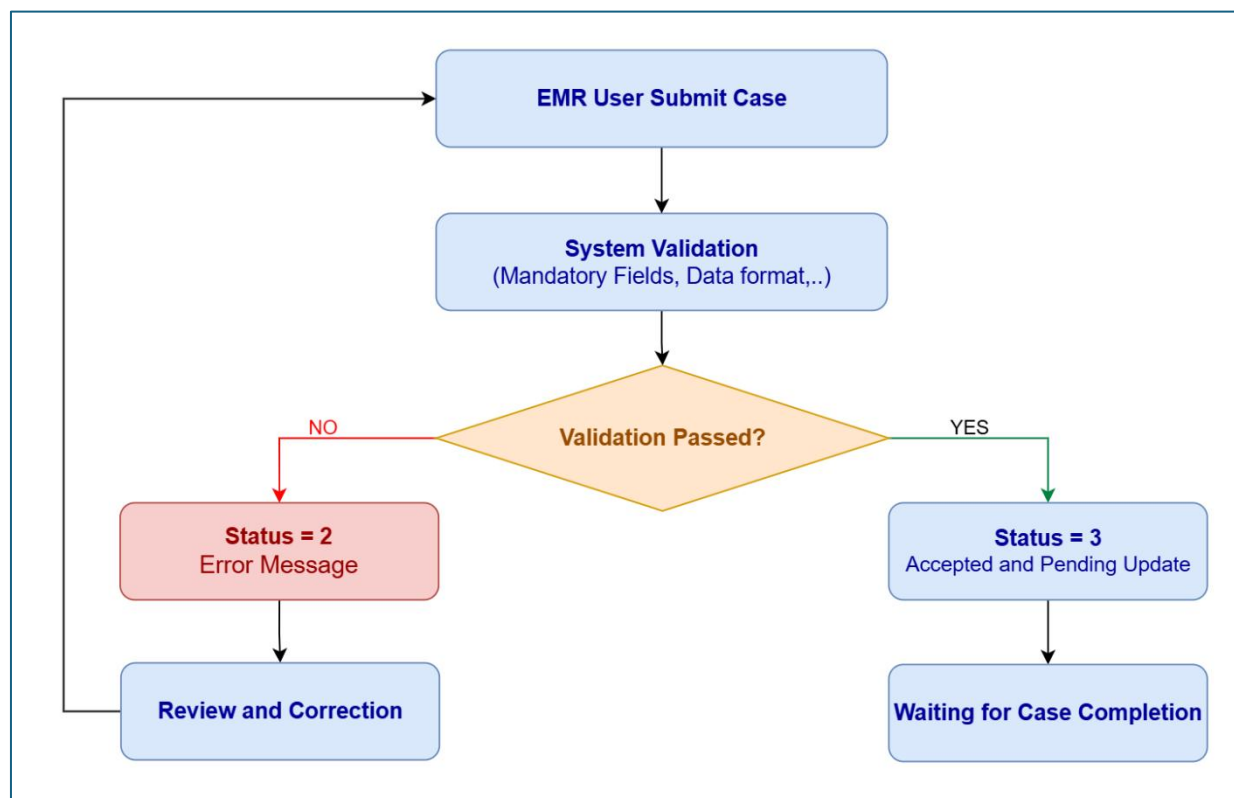
- **Process:** Upon submission, the EMR portal performs local validation and verifies required fields (diagnosis, procedures, etc.).
- **Outcomes:**
 - **Passed:** Request is queued for **NPCA server validation**.
 - **Failed:** Returns **error message** to user with details (e.g., "Missing diagnosis code"). User must correct and resubmit.

3. NPCA Server Validation

After EMR validation passes, the request is sent to the **Digital Payment Certification System**.

- Validates patient eligibility status.
- Performs policy matching (procedure coverage, etc.).
- Generates a unique reference number(ulid) if processed successfully.

2.2.1 Incomplete Case Workflow

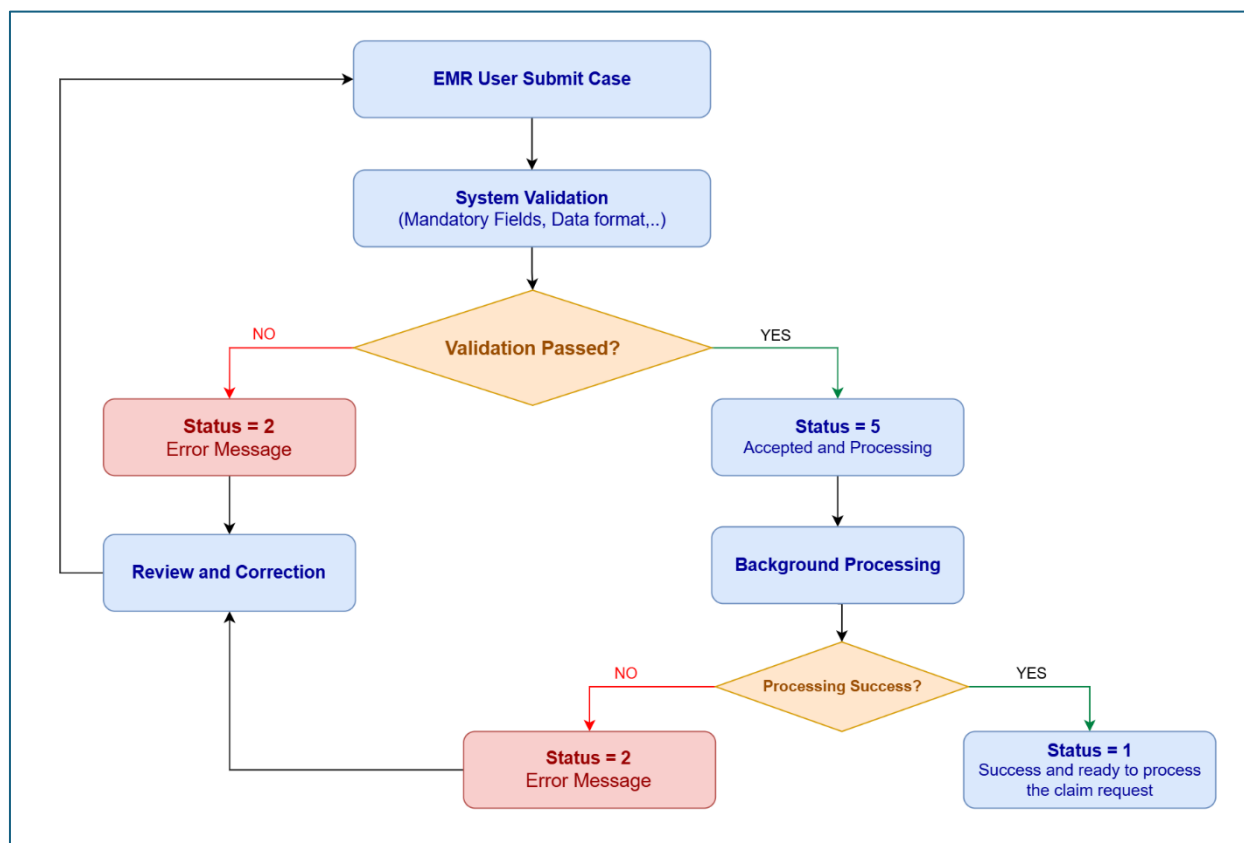


Flow of Incomplete Case

An incomplete case is a visit for which patient care is still in progress. As the treatment has not yet been completed, the visit record remains open and may be updated until the case is finalized.

Status	Meaning	User Action
2	Submission failed	Correct the data and resubmit
3	- The visit has been accepted by the Integration Platform. - The case is not yet complete and requires further updates. - For eligible HEF IPD patients, users may process the initial claim, such as food allowance. - Claim requests cannot be submitted until the visit is completed and updated.	After the patient completes treatment, the healthcare provider must update the visit with the final information and resubmit it to complete the processing workflow. Note: <i>Eligible for HEF food allowance → Healthcare provider submits daily food allowance claims through the Claim Management System (CMS).</i>

2.2.2 Complete Case Workflow



Flow of Complete Case

A complete case refers to a patient visit that has been completed. The patient has finished treatment, and the visit record contains the final information required for processing.

Status	Meaning	User Action
5	Accepted and processing	Wait for the final result
1	Processing completed successfully	The claim request can be initiated
2	Submission failed	Correct the data and resubmit

2.3 COMPONENT ARCHITECTURE

The API follows a modular architecture with interconnected components that represent different aspects of healthcare delivery:

1. **Base Components** - Learn about fundamental structures like Encounters and Observations that serve as building blocks for more complex operations
2. **Core Components** - Essential models such as Patient, Visit, Triage, Treatment, Invoices, and other Clinical that form the core of the healthcare information system
3. **Specific Encounters** - Supporting structures categorized encounters that can easily help identify each case of a patient visit

2.4 DATA MODEL RELATIONSHIPS

The components are interconnected through reference codes:

- Each Patient can have multiple Visits
- Each Visit can include multiple Encounters (Outpatient, Inpatient, Emergency)
- Each Encounter can have associated clinical data (Observations, Diagnoses, Prescriptions, etc.)

2.5 IMPLEMENTATION CONSIDERATIONS

- Designed for interoperability with healthcare standards
- Supports Cambodia-specific requirements (local administrative divisions, facility structures)
- Flexible observation model accommodates various data types

2.6 KEY FEATURES

- Flexible and extensible components
- Comprehensive medical history tracking
- Support for different encounter types (outpatient, inpatient, emergency)
- Integrated billing and payment systems
- Referral management between healthcare facilities
- Detailed clinical documentation with SOAP notes

3 SECURITY & AUTHENTICATION

Parameter	Specification
Protocol	HTTPS (TLS 1.2 or higher)
Authentication	OAuth 2.0 Bearer Token
Required Headers	Authorization: Bearer <access_token> Content-Type: application/json Accept: application/json X-System-Code: <i>[provided by NPCA]</i> X-Health-Facility-Code: <i>[provided by NPCA]</i>

4 API ENDPOINTS

This API provides endpoints to manage patient visits in the healthcare system. It allows submitting patient visits along with related data such as triage information, encounters, diagnoses, prescriptions, and more.

4.1 AUTHENTICATION ENDPOINT

The API uses OAuth 2.0 with Password Client for authentication.

URL	/oauth/token
Method	POST

Request Body	Object: • grant_type: String • client_id: String • client_secret: String • username: String • password: String <i>NPCA Team will provide credential information.</i>
Response	Object: • token_type: String • access_token: String (base 64) • expires_in: Timestamp • refresh_token: String (base 64)

4.2 SEARCH BENEFICIARY

URL	api/v1/external/beneficiary/search
Method	GET
Params	card_number
Response	Objects (can provide beneficiary people): • spid: String • patient_code: String • surname: String • name: String • sex: String • birthdate: YYYY-mm-dd • phone_number: String • nationality: String • relationship_to_head • dead: Boolean

	• death_date: datetime • created_at: datetime • updated_at: datetime • address: object • benefit_cards: array • identifications: array • occupation: string • disabilities: array
--	--

4.3 SUBMISSION ENDPOINT

URL	/api/external/v1/visits
Method	POST
Request Body	<i>Depending on the case and actual treatment procedure, refer to section 4.3.1 and Sample Json[1]</i>
Response	<pre>{ "success": true, "http_code": 200, "message": <i>message</i>, "data": { "status": <i>status_code</i>, "uuld": "<i>unique code generated</i>", "visit_code": "<i>unique code</i>", } }</pre>

4.3.1 Request Body Structure

The following is an overview of the JSON data structured for submission to the endpoint. Each is a component that will be defined later in the document.

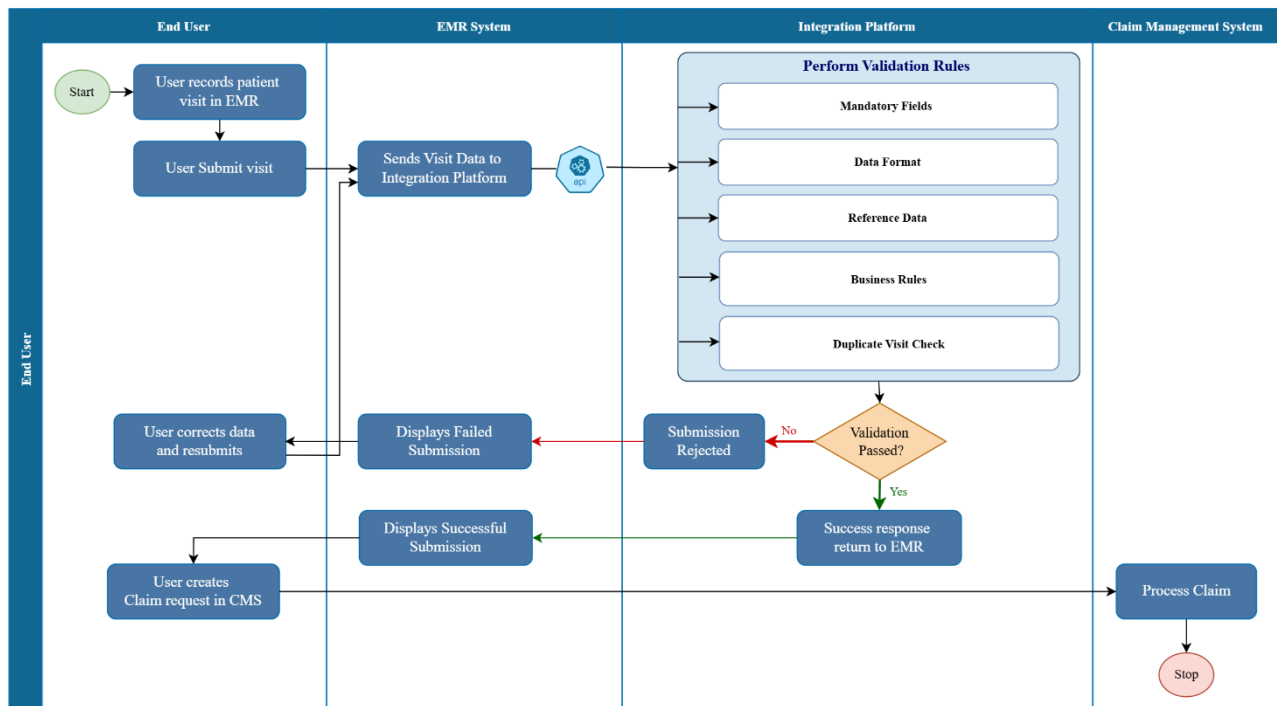
```
{
  "visits": [{
    "patient": {},
```

```

"triages": [{}],
"outpatients": [{}],
"inpatients": [{}],
"emergencies": [{}],
"surgeries": [{}],
"progress_notes ": [{}],
"soaps ": [{}],
"vital_signs": [{}],
"medical_histories": [{}],
"physical_examinations": [{}],
"laboratories": [{}],
"imageries": [{}],
"diagnoses": [{}],
"prescriptions": [{}],
"invoices": [{}],
"referral": {}
}]
}

```

4.3.2 Validation Rules



Flow of Visit Submission Validation and Claim Process

N0	Category	Description
1	Mandatory Fields Validation	The system validates that all required information is provided before accepting a visit submission.
2	Data Format	The system validates all submitted data formats.
3	Reference data	The system validates submitted codes against approved reference data.
4	Business rules	The system validates operational and clinical business rules.

5	Duplicate Visit check	The system checks for duplicate submissions. The same visit must not be submitted multiple times as a new visit.
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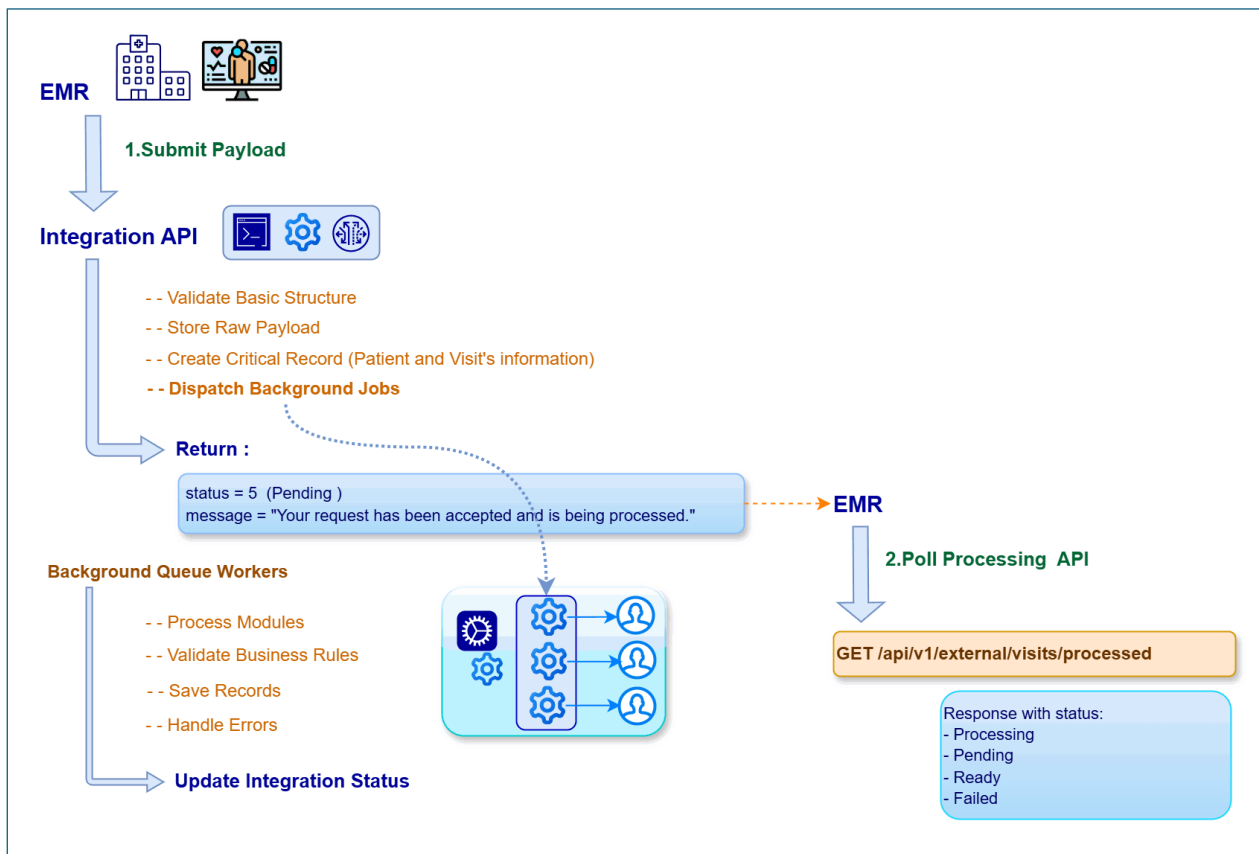
N0	Business Rules Validation	Description
1	Access & Patient Identity Rules	Verify the submitting facility is authorized, and the patient identity is valid and eligible.
2	Basic Visit Rules	Validate visit type, admission and discharge dates, visit status, unique visit code, and other core visit information.
3	Completed Case Rules	Ensure all required clinical and financial information has been submitted, including triage, diagnoses, procedures, invoices, and OPD/IPD-specific data.
4	Clinical Data Linking Rules	Verify relationships between encounters and associated clinical records such as vital signs, SOAP notes, diagnoses, prescriptions, laboratory orders, imaging, physical examinations, and medical history.
5	Resubmission Rules	Validate whether the visit is eligible for resubmission based on its current processing status, admission changes, locked records, pending validation, or previous submission results.

4.4 PROCESSING ENDPOINT

URL	/api/v1/external/visits/processed
Method	GET
Params	- discharged_at_from: YYYY-mm-dd - discharged_at_to: YYYY-mm-dd - admitted_at_from: YYYY-mm-dd - admitted_at_to: YYYY-mm-dd - status: processing / failed / ready / pending - visit_code: String - per_page: Number (Number of items per page) <i>Importance: Apply this enhancement to improve the identification of records that require updates.</i>
Response	Object - success: Boolean - http_code: Number - message: String - data: Objects

4.5 TRACKING PROCESSING RESULT

4.5.1 Submission API Flow



Flow of Submission Tracking Process

4.5.2 Implement Processing Status Retrieval

EMR systems must retrieve processing results after a visit enters Status 5. This ensures users receive final processing outcomes.

Benefits:

- Automatic update on EMR UI
- No manual checking
- Better user experience

4.5.3 Display Processing Status

Users should be able to see:

- Processing
- Failed
- Ready
- Pending

directly from the EMR interface.

4.5.4 Submission Status Reference

Status	Processing Status	Meaning
5	Processing	Background processing in progress
3	Pending	Additional updates are required before processing can continue.
2	Failed	Validation or processing failure
1	Ready	Processing completed successfully

4.5.5 Display Failure Reasons

When processing fails, the EMR system should display the reason for the failure returned by the Integration Platform.

Examples:

- Referral reason is required.
- Diagnosis code is invalid.
- Visit Code has been submitted.
- Systolic Blood Pressure must be a number.

4.5.6 Support Data Correction and Resubmission

When processing fails:

1. User reviews the failure reason.
2. User corrects the data.
3. User resubmits the visit.

The EMR system must support this workflow.

4.6 SUBMISSION RESPONSE HANDLING

4.6.1 Success Responses

Response	Description
<pre>{ "success": true, "http_code": 200, "message": "Your request is accepted and pending update.", "data": { "visit_ulid": "{ ulid }", "visit_code": "{ visit_code }", "status": 3 } }</pre>	Submission has been accepted, and additional updates are required before processing can continue.
<pre>{ "success": true,</pre>	Submission has been accepted and is being processed in the background.

<pre> "http_code": 200, "message": "Your request is accepted and is being processed.", "data": { "visit_ulid": "{ ulid }", "visit_code": "{ visit_code }", "status": 5 } } </pre>	
---	--

4.6.2 Error Response

4.6.2.1 400 Bad Request

```

{
  "success": false,
  "http_code": 400,
  "message": "Invalid request data",
  "errors": "Error Message"
}

```

4.6.2.2 401 Unauthorized

```

{
  "success": false,
  "http_code": 401,
  "message": "Unauthenticated",
  "errors": "Invalid or expired token"
}

```

4.6.2.3 403 Forbidden

```

{
  "success": false,
  "http_code": 403,
  "message": "Access denied",
  "errors": "You do not have permission to create visits"
}

```

4.6.2.4 500 Server Error

```

{
  "success": false,
  "http_code": 403,
  "message": "Internal server error",
  "errors": "An unexpected error occurred while processing your request"
}

```

4.6.2.5 Validation Error

Format	Example
{	{

<pre> "success": false, "http_code": 422, "message": "Invalid request data", "errors": "{ error_field_name } { error_message }", "data": null } </pre>	<pre> "success": false, "http_code": 422, "message": "Invalid request data", "errors": "The visits.0.code field is required.", "data": null } </pre>
--	--

5 BASE COMPONENTS

This section attempts to explain the basic concept of each component used under the visit component.

5.1 ENCOUNTER

An Encounter represents a clinical interaction between a patient and a healthcare provider during a visit.

5.1.1 Data Model

Field	Type	Required	Description
patient_code	string	YES	Reference to the patient's unique code
visit_code	string	YES	Reference to visit the unique code
code	string	YES	Unique identifier for the encounter
class	string	YES	Class of encounter ("IPD", "OPD", or "EMERGENCY")
name	string	YES	What is the encounter (triage, ward names, sub-clinic ...)
service_type	string	YES	Type of the service provided (nullable)
started_at	datetime	YES	Date and time encounter started
ended_at	datetime	YES	Date and time encounter ended (nullable)
encountered_by	string	YES	Name of healthcare provider
title	string	YES	Whether the healthcare provider is doctor or nurse
created_at	datetime	YES	Record creation timestamp
updated_at	datetime		Record the last update timestamp

5.1.2 Example

```

{
  "patient_code": "P25003877",
  "visit_code": "V420650",
  "code": "E317001",
  "class_name": "OPD",
  "name": "General Medicine"
  "service_type": "General Consultation",
  "started_at": "2025-03-17T09:35:00.000+07:00",
  "ended_at": "2025-03-17T10:15:00.000+07:00",
  "encountered_by": "Dr. Chhun Sovannarith"
  "title": "Doctor",

```

```

"created_at": "2025-03-17T09:35:00.000+07:00",
"updated_at": "2025-03-17T10:15:00.000+07:00"
}

```

5.2 OBSERVATIONS

Observations contain clinical data about a patient. They are designed to be flexible enough to capture virtually any type of patient information.

5.2.1 Data Model

Field	Type	Required	Description
encounter_code	string	YES	Reference to encounter unique code
observation_code	string		Reference to parent observation (nullable)
name	string	YES	Name for the observation
category	string	YES	Name for the category of the observation if applicable
value	various	YES	Value of the observation (type depends on value_type)
value_type	string	YES	Type of value ("boolean", "string", "text", "integer", "float", "datetime", "time", "array", "complex")
value_unit	string		Unit of measurement (nullable)

5.2.2 Value Types

The observation structure needs to be flexible so any type of data stored in the EMR can be submitted via API. To introduce flexibility, a value type is necessary. It basically determines what type of data it is in the value field.

5.2.2.1 Boolean Example

```

{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Smoker",
  "category": "Social History",
  "value": false,
  "value_type": "boolean",
  "value_unit": null
}

```

5.2.2.2 String Example

```

{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Blood Type",
  "category": "Laboratory",
  "value": "O+",
  "value_type": "string",
  "value_unit": null
}

```

5.2.2.3 Text Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Clinical Notes",
  "category": "Assessment",
  "value": "Patient reports symptoms of malaria including periodic
fever, headache, and fatigue. Has history of travel to Mondulkiri
province two weeks ago.",
  "value_type": "text",
  "value_unit": null
}
```

5.2.2.4 Numeric Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Blood Glucose",
  "category": "Vital Signs",
  "value": 110,
  "value_type": "integer",
  "value_unit": "mg/dL"
}
```

5.2.2.5 Datetime Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Last Meal Time",
  "category": "Nutrition",
  "value": "2025-03-17T07:00:00.000+07:00",
  "value_type": "datetime",
  "value_unit": null
}
```

5.2.2.6 Time Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Duration of Symptoms",
  "category": "History",
  "value": "48:00:00",
  "value_type": "time",
  "value_unit": "hh:mm:ss"
}
```

5.2.2.7 Array Example

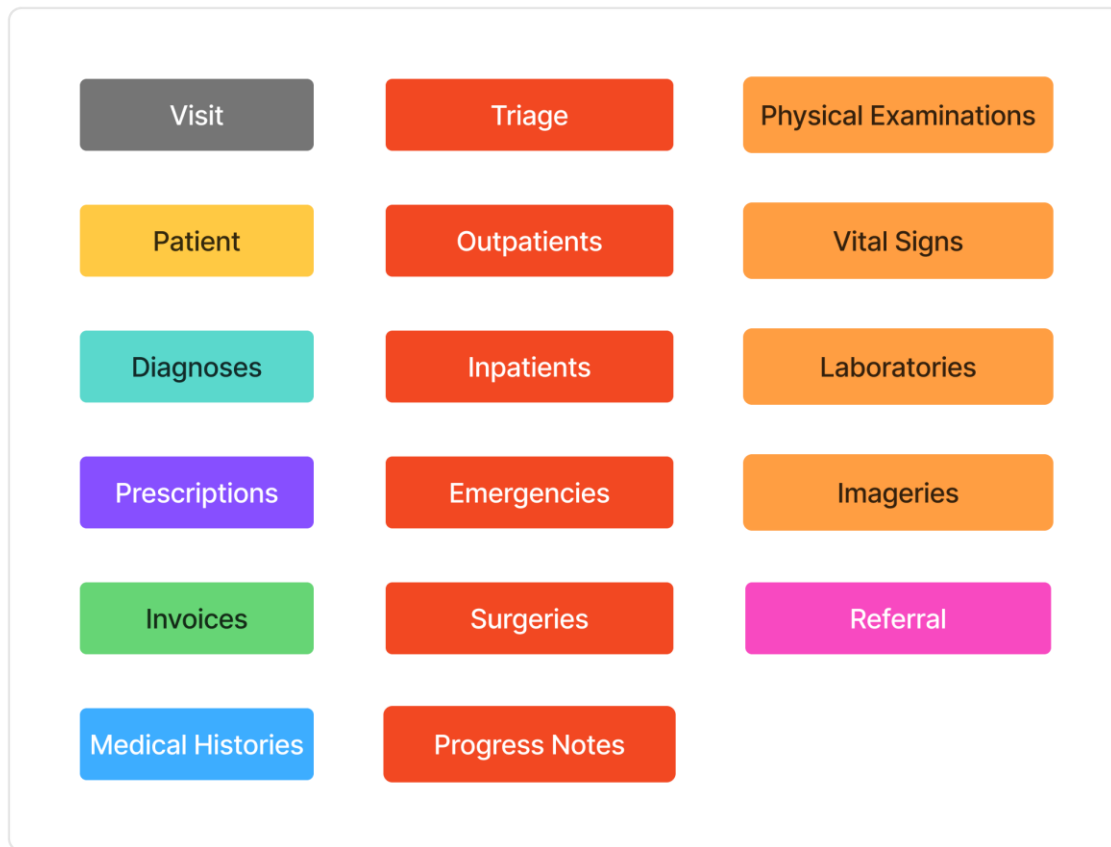
```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Symptoms",
  "category": "History",
  "value": ["Fever", "Headache", "Fatigue", "Joint Pain"],
  "value_type": "array",
  "value_unit": null
}
```

5.2.2.8 Complex Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Malaria Risk Assessment",
  "category": "Risk Assessment",
  "value": {
    "risk_level": "High",
    "travel_history": true,
    "prevention_measures": ["Bed nets", "Repellent"],
    "endemic_area": "Mondulkiri province"
  },
  "value_type": "complex",
  "value_unit": null
}
```

Note: conditions also have the same structure and values as observations. But for simplicity, the conditions in the Medical History components are shortened and unified to make it easier to implement them.

6 CORE COMPONENTS



6.1 PATIENT

The Patient object contains personal information and demographics for an individual under medical care.

6.1.1 Data Model

Field	Type	Required	Description
code	string	YES	Unique identifier for the patient in the system
surname	string	YES	Patient's family name
name	string	YES	Patient's given name
sex	string	YES	Patient's biological sex
birthdate	date	YES	Patient's date of birth
phone	string	YES	Patient's contact number
nationality	string	YES	Patient's nationality
disabilities	array of strings		List of disabilities, if applicable
occupation	string		Patient's job or profession (nullable)
marital_status	string	YES	Patient's marital status (nullable)
photos	array of strings		URLs to patient photos (nullable)

address	object	YES	Patient's address information (see Address model)
identifications	array of objects		Patient identification documents (see Identification model)
death_date	datetime		Date and time of death, if applicable
spid	string	YES	Patient's Social Protection ID
created_at	datetime	YES	Record creation timestamp
updated_at	datetime	YES	Record the last update timestamp

6.1.2 Address Model

Field	Type	Required	Description
province	object	YES	Information about the province
province.code	string	YES	Province code from NCDD standard
province.name	string	YES	Province's name in Khmer
district	object	YES	Information about the district
district.code	string	YES	District code from NCDD standard
district.name	string	YES	District name in Khmer
commune	object	YES	Information about the commune
commune.code	string	YES	Commune code from NCDD standard
commune.name	string	YES	Commune name in Khmer
village	object	YES	Information about the village
village.code	string	YES	Village code from NCDD standard
village.name	string	YES	Village name in Khmer
house_number	string		House number (nullable)
street_number	string		Street number (nullable)
location	string		Additional location details (nullable)

6.1.3 Identification Model

Field	Type	Required	Description
patient_code	string	YES	Reference to patient unique code
card_code	string	YES	Identification number
card_type	string	YES	Type of identification document

6.1.4 Example

```
{
  "patient": {
    "code": "P25003877",
    "surname": "សៅ",
    "name": "សុភ័ន្ទ",
    "sex": "M",
```

```

"birthdate": "1980-05-15",
"phone": "012 345 678",
"nationality": "Cambodian",
"disabilities": ["Hearing impairment"],
"occupation": "Teacher",
"marital_status": "Married",
"photos": ["https://hospital.example/photos/PT20250317001_1.jpg"],
"address": {
  "province": {
    "code": "12",
    "name": "Phnom Penh"
  },
  "district": {
    "code": "1201",
    "name": "Chamkar Mon"
  },
  "commune": {
    "code": "120101",
    "name": "Tonle Bassac"
  },
  "village": {
    "code": "12010101",
    "name": "Phsar Deum Thkov"
  },
  "house_number": "45",
  "street_number": "302",
  "location": "Near Central Market"
},
"identifications": [
  {
    "patient_code": "P25003877",
    "card_code": "ID123456789",
    "card_type": "National ID"
  }
],
"death_date": null,
"spid": "250317001",
"created_at": "2025-03-17T08:30:00.000+07:00",
"updated_at": "2025-03-17T08:30:00.000+07:00"
}

```

6.2 VISIT

A Visit represents a patient's stay at a healthcare facility, from admission to discharge.

6.2.1 Data Model

Field	Type	Required	Description
health_facility_code	string	YES	Unique identifier for the healthcare facility, issued by NPCA

patient_code	string	YES	Reference to the patient's unique code
code	string	YES	Unique identifier for the visit
admission_type	string	YES	How a patient is admitted to a healthcare facility
discharge_type	string	YES	How a patient is discharged from a healthcare facility
visit_outcome	string	YES	The patient's condition at the end of the visit
visit_type	string	YES	The type or purpose of a patient's visit (IPD or OPD)
admitted_at	datetime	YES	Date and time of admission
discharged_at	datetime	YES	Date and time of discharge
followup_at	datetime		When the patient is scheduled to return for follow-up care
created_at	datetime	YES	Record creation timestamp
updated_at	datetime	YES	Record the last update timestamp

6.2.2 Example

```
{
  "visits": [
    {
      "health_facility_code": "121020",
      "patient_code": "P25003877",
      "code": "V420650",
      "admission_type": "Walk-in",
      "discharge_type": "Authorized",
      "visit_outcome": "Improved",
      "visit_type": "OPD",
      "admitted_at": "2025-03-17T09:30:00.000+07:00",
      "discharged_at": "2025-03-17T10:45:00.000+07:00",
      "followup_at": "2025-03-24T09:00:00.000+07:00",
      "created_at": "2025-03-17T09:30:00.000+07:00",
      "updated_at": "2025-03-17T10:45:00.000+07:00"
    }
    ...
  ]
}
```

6.3 TRIAGE

Triage represents the initial assessment of a patient to determine treatment priority.

6.3.1 Data Model

Field	Type	Required	Description
patient_code	string	YES	Reference to patient's unique code
visit_code	string	YES	Reference to visit the unique code
encounter_code	string	YES	Reference to encounter a unique code (nullable)
code	string	YES	Unique identifier for the triage

chief_complaint	text	YES	Primary symptom or problem reported by the patient at the time of the visit
height	number	YES	Value of height of the patient in CM
weight	number	YES	Value of weight of the patient in KG
recorded_at	datetime	YES	Date and time triage was recorded
recorded_by	string	YES	Name of the healthcare provider who recorded the triage
title	string		Position of the healthcare provider

6.3.2 Example

```
{
  "triages": [{
    "visit_code": " V420650",
    "encounter_code": "P25003877",
    "code": "TR250317",
    "chief_complaint": "Periodic fever with chills for 3 days",
    "height": 170,
    "weight": 65,
    "recorded_at": "2025-03-17T09:32:00.000+07:00",
    "recorded_by": "Khun Srey Mom",
    "title": "Doctor",
  }]
}
```

6.4 VITAL SIGNS

Vital signs contain key physiological measurements of a patient. This object must contain all the vital signs information, such as temperature, pulse (heart rate), respiratory rate, systolic blood pressure, diastolic blood pressure, oxygen saturation, and an optional random rapid blood glucose test.

Vital Sign	Normal Range	Units
Temperature	36.5–37.3 °C	°C
Pulse (Heart Rate)	60–100 beats per minute	/mn
Respiratory Rate	12–20 breaths per minute	/mn
Blood Pressure	Systolic: 90–120 mmHg, Diastolic: 60–80 mmHg	mmHg
Oxygen Saturation (SpO ₂)	95–100%	%
Random Glucose Rapid Test	Random: < 140 mg/dL, Fasting: 70–99 mg/dL	mg/dL

6.4.1 Data Model

Field	Type	Required	Description
encounter_code	string	YES	The code of the patient encounter when the vital signs were recorded. It can match different encounter types depending on the context: Triage.code – if recorded during triage

			• IPD.code – if recorded during an inpatient visit • Emergency.code – if recorded during an emergency visit • ProgressNote.code – if recorded during a routine daily assessment
recorded_at	datetime	YES	Date and time vital signs were recorded
recorded_by	string	YES	Name of healthcare provider who recorded the vital signs
title	string	YES	Doctor or nurse
observations	array of objects	YES	Array of vital sign measurements

6.4.2 Observation in Vital Sign

Field	Type	Required	Description
name	string	YES	Name of the vital sign
value	numeric	YES	Measured value

6.4.3 Example

```
{
  "vital_signs": [
    {
      "encounter_code": "E417001",
      "recorded_at": "2025-03-17T09:40:00.000+07:00",
      "recorded_by": "Khun Srey Mom",
      "title": "Doctor",
      "observations": [
        {
          "name": "Systolic Blood Pressure",
          "value": 135,
        },
        {
          "name": "Diastolic Blood Pressure",
          "value": 85,
        },
        {
          "name": "Heart Rate",
          "value": 92,
        },
        {
          "name": "Respiratory Rate",
          "value": 18,
        },
        {
          "name": "Temperature",
          "value": 38.7,
        },
      ],
    },
  ],
}
```

```

    "name": "Blood Oxygen",
    "value": 97,
  },
  {
    "name": "Random Blood Glucose",
    "value": 110,
  }
]
}
]
}

```

6.5 MEDICAL HISTORIES

Medical History records the patient's past and current health conditions for clinical reference.

6.5.1 Components

Category	Section	Description
Current Illness	History of Present Illness	Detailed narrative of symptoms — onset, duration, severity, aggravating/relieving factors, associated symptoms
Current Illness	Current Medication	Current medications (name, dose, frequency), over-the-counter drugs, supplements
Health History	Past Medical History	Previous diagnoses (e.g., diabetes, hypertension), hospitalizations, chronic conditions
Health History	Past Surgical History	Details of past surgeries (type, date, complications, if any)
Health History	Allergies	Known food, drug, or environmental reaction
Health History	Immunization	Vaccination status — especially for tetanus, flu, COVID-19, and childhood vaccines
Women's Health	Obstetric & Gynecologic History	For female patients: menarche, menstruation pattern, pregnancies, contraception, and menopause
Family & Lifestyle	Family History	Hereditary conditions in parents, siblings, and grandparents. Example: heart disease, cancer
Family & Lifestyle	Social History	Lifestyle factors: tobacco use, alcohol, drug use, occupation, living situation, marital status

Note: the medical history detailed information must use these section names.

6.5.2 Data Model

Field	Type	Required	Description
medical_history	array of conditions objects	NO	Array of conditions (nullable)

6.5.3 Condition Model

Field	Type	Required	Description
patient_code	string	YES	Reference to the patient unique code
visit_code	string	YES	Reference to visit the unique code
encounter_code	text	YES	Reference to encounter a unique code
name	string	YES	Name of the condition
value	array of string	YES	Array of value in string of the condition

6.5.4 Example

```
{
  "medical_histories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Immunization History",
      "value": [
        "BCG",
        "OPV",
        "DTP",
        "Measles",
        "COVID-19"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Allergies",
      "value": [
        "Prahok",
        "Shrimp paste",
        "Penicillin"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Past Surgical History",
      "value": [
        "Appendectomy Surgery"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Past Medical History",

```

```

    "value": [
      "Dengue fever (2020)",
      "Typhoid (2018)",
      "Cesarean section at Calmette Hospital (2019)"
    ]
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Family History",
    "value": [
      "Mother with diabetes",
      "Father with hypertension and stroke at age 60"
    ]
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Current Illness History",
    "value": [
      "Patient reports high fever with chills occurring every 48
hours, accompanied by headache and joint pain. Symptoms started 3 days
ago after returning from a trip to Mondulkiri province. No prior history
of malaria. No medication taken yet."
    ]
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Current Medication",
    "value": [
      "Metformin 500mg",
      "Amlodipine 5mg"
    ]
  }
]
}

```

*Note: if the value is empty or not available, the object must not be included

6.6 PHYSICAL EXAMINATIONS

Physical Examination contains clinical findings from examination of a patient.

6.6.1 Assessment Points

System/Section	Assessment Points
General Appearance	Consciousness (alert/drowsy), Distress, Nutrition, Posture, Gait, Hygiene, Pallor, Jaundice, Cyanosis, Edema

Glasgow Coma Scale	Eye opening, Verbal response, Motor response
Head	Scalp condition, Hair distribution, Lesions, Deformities
Eyes	Pupils (size, reaction), Conjunctiva (pallor), Sclera (icterus), Eye movements
Ears	Canal and Tympanic membrane, Discharge, Hearing
Nose	Nasal mucosa, Septum deviation, Discharge, Polyps
Throat/Mouth	Tonsils, Pharynx, Uvula, Oral mucosa, Tongue, Teeth
HEENT*	Information of Head, Eyes, Ears, Nose, Throat can be combined
Neck	Lymph nodes, Thyroid gland, Tracheal position, JVP
Cardiovascular System	Inspection (chest), Palpation (apex beat), Auscultation (S1/S2, murmurs), Peripheral pulses
Respiratory System	Chest symmetry, Expansion, Percussion (dullness), Breath sounds, Crepitations, Wheezing
Digestive System	Inspection (distension), Palpation (tenderness, organ size), Percussion, Bowel sounds
Genitourinary System	External genitalia (if applicable), Bladder palpation, Costovertebral angle tenderness, Discharge (if present)
Musculoskeletal System	Joint swelling, Deformity, Range of motion, Muscle tone/strength, Spine, Gait
Nervous System	Mental status, Cranial nerves, Motor/sensory function, Reflexes, Coordination
Skin	Rashes, Lesions, Color changes, Ulcers, Turgor, Temperature
Lymph Nodes	Cervical, Axillary, Inguinal – enlargement, tenderness, mobility

6.6.2 Data Model

Field	Type	Required	Description
physical_examinations	array of object	NO	Array of assessment points (nullable)

6.6.3 Physical Examination Model

Field	Type	Required	Description
patient_code	string	YES	Reference to patient unique code
visit_code	string	YES	Reference to visit unique code
encounter_code	string		Reference to encounter unique code
name	string	YES	Name for the physical assessment point
value	various	YES	Value of the assessment result
value_type	string	YES	Type of Value
value_unit	string		Measurement unit represent the value

6.6.4 Example

```
{
  "physical_examinations": [
    {
```

```

    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "General Appearance",
    "value": "Ill-appearing with intermittent shivering",
    "value_type": "text",
    "value_unit": null
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Skin",
    "value": "Warm, jaundiced, no rash",
    "value_type": "text",
    "value_unit": null
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E317001",
    "name": "Digestive System",
    "value": [
      {
        "inspection": "Flat, no visible masses"
      },
      {
        "palpation": "Soft, mild RUQ tenderness, spleen palpable 1cm
below costal margin"
      },
      {
        "percussion": "No shifting dullness"
      },
      {
        "auscultation": "Normal bowel sounds"
      }
    ],
    "value_type": "complex",
    "value_unit": null
  }
]
}

```

6.7 LABORATORIES

6.7.1 Data Model

Field	Type	Required	Description
laboratories	array of object	NO	Array of laboratory tests (nullable)

6.7.2 Laboratories Model

Field	Type	Required	Description
patient_code	string	YES	Reference to patient unique code
visit_code	string	YES	Reference to visit unique code
encounter_code	string		Reference to encounter unique code
srequest_code	string	YES	Unique code represents the lab request
requested_at	datetime	YES	Date and time laboratory were requested
requested_by	string	YES	Name of doctor who requested the laboratory
title	string		Name of doctor
results	array of object	YES	Array of laboratory objects

6.7.3 Result Model

Field	Type	Required	Description
name	string	YES	The item of lab result
category	string	YES	Category that lab result fall under (e.g. biochemistry, hematology)
value	various	YES	Measured value
value_type	string	YES	Type of value
value_unit	string	YES	Unit of measurement
reference_range	string	YES	Normal range for the measurement
interpretation	string	YES	Interpretation of the result
verified_at	datetime	YES	Date and time of doctor who verified
verified_by	string	YES	Name of healthcare provider who verified
recorded_at	datetime	YES	Date and time of doctor who recorded
recorded_by	string	YESs	Name of healthcare provider who recorded

6.7.4 Example – Structured

```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-03-17T10:30:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "title": "Doctor",
      "results": [
        {
          "name": "Malaria Blood Smear",
          "category": "Parasitology",
          "value": "Positive for P. falciparum",
          "value_type": "string",
```

```

        "value_unit": null,
        "reference_range": "Negative",
        "interpretation": "Positive",
        "verified_at": "2025-03-17T10:50:00.000+07:00",
        "verified_by": "Dr. Meas Chenda",
        "recorded_at": "2025-03-17T10:52:00.000+07:00",
        "recorded_by": "Meas Chandaravuth"
    },
    {
        "name": "Parasitemia",
        "category": "Parasitology",
        "value": 2.5,
        "value_type": "float",
        "value_unit": "%",
        "reference_range": "0",
        "interpretation": "High",
        "verified_at": "2025-03-17T10:50:00.000+07:00",
        "verified_by": "Dr. Meas Chenda",
        "recorded_at": "2025-03-17T10:52:00.000+07:00",
        "recorded_by": "Meas Chandaravuth"
    },
    {
        "name": "Hemoglobin",
        "category": "Hematology",
        "value": 11.2,
        "value_type": "float",
        "value_unit": "g/dL",
        "reference_range": "13.5-17.5",
        "interpretation": "Low",
        "verified_at": "2025-03-17T10:50:00.000+07:00",
        "verified_by": "Dr. Meas Chenda",
        "recorded_at": "2025-03-17T10:52:00.000+07:00",
        "recorded_by": "Meas Chandaravuth"
    }
]
}

```

6.7.5 Example - Complex

```

{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-04-22T09:15:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "results": [
        {
          "name": "Complete Blood Count",

```

```

"category": "Hematology",
"value": {
  "Hemoglobin": {
    "result": "11.2 g/dL",
    "reference": "13.5-17.5 g/dL",
    "interpretation": "Low"
  },
  "White Blood Cells": {
    "result": "4.5 x10^3/μL",
    "reference": "4.5-11.0 x10^3/μL",
    "interpretation": "Normal"
  },
  "Red Blood Cells": {
    "result": "4.1 x10^6/μL",
    "reference": "4.5-5.9 x10^6/μL",
    "interpretation": "Low"
  },
  "Platelets": {
    "result": "120 x10^3/μL",
    "reference": "150-450 x10^3/μL",
    "interpretation": "Low"
  },
  "Hematocrit": {
    "result": "34%",
    "reference": "41-50%",
    "interpretation": "Low"
  },
  "MCV": {
    "result": "83 fL",
    "reference": "80-96 fL",
    "interpretation": "Normal"
  },
  "MCH": {
    "result": "27.3 pg",
    "reference": "27-33 pg",
    "interpretation": "Normal"
  },
  "MCHC": {
    "result": "32.9 g/dL",
    "reference": "32-36 g/dL",
    "interpretation": "Normal"
  }
},
"value_type": "object",
"value_unit": null,
"reference_range": null,
"interpretation": null,
"verified_at": "2025-03-17T10:50:00.000+07:00",
"verified_by": "Dr. Meas Chenda",
"recorded_at": "2025-03-17T10:52:00.000+07:00",
"recorded_by": "Meas Chandaravuth"
},

```

```

{
  "name": "Liver Function",
  "category": "Chemistry",
  "value": {
    "ALT": {
      "result": "45 U/L",
      "reference": "7-56 U/L",
      "interpretation": "Normal"
    },
    "AST": {
      "result": "42 U/L",
      "reference": "10-40 U/L",
      "interpretation": "High"
    },
    "ALP": {
      "result": "110 U/L",
      "reference": "44-147 U/L",
      "interpretation": "Normal"
    },
    "Total Bilirubin": {
      "result": "1.2 mg/dL",
      "reference": "0.1-1.2 mg/dL",
      "interpretation": "High-normal"
    },
    "Direct Bilirubin": {
      "result": "0.3 mg/dL",
      "reference": "0.0-0.3 mg/dL",
      "interpretation": "High-normal"
    },
    "Total Protein": {
      "result": "7.2 g/dL",
      "reference": "6.0-8.3 g/dL",
      "interpretation": "Normal"
    },
    "Albumin": {
      "result": "4.0 g/dL",
      "reference": "3.4-5.4 g/dL",
      "interpretation": "Normal"
    }
  },
  "value_type": "object",
  "value_unit": null,
  "reference_range": null,
  "interpretation": null,
  "verified_at": "2025-03-17T10:50:00.000+07:00",
  "verified_by": "Dr. Meas Chenda",
  "recorded_at": "2025-03-17T10:52:00.000+07:00",
  "recorded_by": "Meas Chandaravuth"
}
]
}

```

```
]
}
```

6.7.6 Example – File

```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-04-22T11:00:00.000+07:00",
      "requested_by": "Dr. Tep Veasna",
      "results": [
        {
          "name": "Blood Smear",
          "category": "Parasitology",
          "value":
["https://example.com/lab/EN20250422002/blood_smear_field1.jpg","https://example.com/lab/EN20250422002/blood_smear_field2.jpg","https://example.com/lab/EN20250422002/blood_smear_field3.jpg"]
          ,
          "value_type": "array",
          "value_unit": null,
          "reference_range": null,
          "interpretation": "Positive for P. falciparum",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        },
        {
          "name": "Microbiology Culture",
          "category": "Microbiology",
          "value":
["https://example.com/lab/EN20250422002/culture_plate1.jpg","https://example.com/lab/EN20250422002/culture_plate2.jpg","https://example.com/lab/EN20250422002/gram_stain.jpg","https://example.com/lab/EN20250422002/sensitivity_test.jpg"],
          "value_type": "array",
          "value_unit": null,
          "reference_range": null,
          "interpretation": "Positive for Salmonella typhi",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        }
      ]
    }
  ]
}
```

```
]
}
```

6.8 IMAGERIES

6.8.1 Data Model

Field	Type	Required	Description
imageries	array of object	NO	Array of medical imaging records for a patient encounter (nullable)

6.8.2 Imageries Model

Field	Type	Required	Description
patient_code	string	YES	Reference to patient's unique code
visit_code	string	YES	Reference to visit the unique code
encounter_code	string		Reference to encounter a unique code
request_code	string	YES	Unique code represents the lab request
requested_at	datetime	YES	Date and time of the imagery requested
requested_by	string	YES	Name of the healthcare provider who requested the imagery
title	string	YES	Name of doctor
results	array of object	YES	Array of imagery objects

6.8.3 Result Model

Field	Type	Required	Description
name	string	YES	The item of the imaging result
category	string	YES	Category that imaging falls under (e.g. radiology, ...)
images	array of string	YES	Measured value
result	string	YES	Full result of the imagery
conclusion	string	YES	Summary of the result
verified_at	datetime	YES	Date and time that the sample was verified
verified_by	string	YES	Name of healthcare provider who verified the sample
recorded_at	datetime	YES	Date and time results were recorded
recorded_by	String	YES	Name of the healthcare provider who recorded the result

6.8.4 Example

```
{
  "imageries": [
    {
      "patient_code": "P25003877",
```

```

"visit_code": "V420650",
"encounter_code": "E417001",
"request_code": "IM2301",
"requested_at": "2025-03-17T11:30:00.000+07:00",
"requested_by": "Dr. Chhun Sovannarith",
"title": "title",
"results": [
  {
    "name": "Brain Scan",
    "category": "CT Scan",
    "images": [
      "img.com/result2.jpg"
    ],
    "result": "string",
    "conclusion": null,
    "verified_at": "2025-03-17T10:50:00.000+07:00",
    "verified_by": "Dr. Meas Chenda",
    "recorded_at": "2025-03-17T10:52:00.000+07:00",
    "recorded_by": "Meas Chandaravuth"
  }
],
},
{
  "patient_code": "P25003877",
  "visit_code": "V420650",
  "encounter_code": "E417001",
  "requested_at": "2025-03-17T11:35:00.000+07:00",
  "requested_by": "Dr. Chhun Sovannarith",
  "title": "Doctor",
  "results": [
    {
      "name": "Adomen Ultrasound",
      "category": "Ultrasound",
      "images": [
        "img.com/result1.jpg"
      ],
      "result": "string",
      "conclusion": null,
      "verified_at": "2025-03-17T10:50:00.000+07:00",
      "verified_by": "Dr. Meas Chenda",
      "recorded_at": "2025-03-17T10:52:00.000+07:00",
      "recorded_by": "Meas Chandaravuth"
    }
  ]
}
]
}

```

6.9 DIAGNOSIS

6.9.1 Diagnosis Types

Diagnosis Type	Description
Differential	A list of possible conditions that could explain the symptoms
In	The condition is suspected when the patient is first admitted
Out	Final diagnosis after treatment and evaluation, recorded at discharge
Primary	The main reason for the current hospital visit or treatment
Secondary	Coexisting conditions that influence care or outcomes

6.9.2 Diagnosis Model

Field	Type	Require	Description
patient_code	string	YES	Reference to patient unique code
visit_code	string	YES	Reference to visit unique code
encounter_code	string		Reference to encounter unique code
diagnosis_type	string	YES	Name of diagnosis type as above table
diagnosis_code	string	YES	ICD-10 code for the diagnosis
diagnosis_name	string	YES	Name of the diagnosis
diagnosis_description	string		Detailed description of the diagnosis
diagnosed_at	datetime	YES	Date and time of diagnosis
diagnosed_by	string	YES	Name of healthcare provider who made the diagnosis
title	string		Position of the healthcare provider (doctor or nurse)

6.9.3 Example

```
{
  "diagnosis": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E317001",
      "diagnosis_type": "Primary",
      "diagnosis_code": "B50",
      "diagnosis_name": "Plasmodium falciparum malaria",
      "diagnosis_description": "Plasmodium falciparum malaria with cerebral complications",
      "diagnosed_at": "2025-03-17T10:05:00.000+07:00",
      "diagnosed_by": "Dr. Chhun Sovannarith",
      "title": "Doctor"
    }
  ]
}
```

6.10 SOAPS

SOAPS are structured clinical notes that record Subjective, Objective, Assessment, and Plan information. They may supplement detailed clinical data or provide a summary of the patient's clinical information

6.10.1 Data Model

Field	Type	Require	Description
soaps	array of object	NO	Array of clinical notes written using the SOAP format for a patient encounter (nullable)

6.10.2 SOAPS Model

Field	Type	Require	Description
encounter_code	string	YES	Reference to encounter unique code
subjective	text		Patient's reported symptoms and feelings
objective	text		Provider's objective
assessment	text		Provider's assessment of the patient situation
evaluation	text		Provider's summary or evaluation regarding the assessment
plan	text		Plan for treatment or next steps
code	string		Unique identifier for the SOAPS
recorded_at	datetime		Date and time SOAPS was written
recorded_by	string		Name of healthcare provider who wrote the SOAPS
title	string		Position of the healthcare provider (doctor or nurse)

6.10.3 Example

```
{
  "soaps": [
    {
      "encounter_code": "EN20250317001",
      "code": "SP3177998",
      "subjective": "Patient reports fever has decreased after first dose of medication. Still experiencing mild headache and fatigue. No chills since this morning. Able to tolerate oral intake well.",
      "objective": "Temperature 37.8°C (down from 38.7°C). Heart rate 85/min. Blood pressure 125/80 mmHg. Skin less jaundice. Abdomen remains soft with mild tenderness in RUQ.",
      "assessment": "Uncomplicated malaria showing initial response to treatment. Fever is trending down. No signs of complications.",
      "plan": "Continue artemether-lumefantrine as prescribed. Encourage oral hydration. Follow-up in clinic in 3 days. Return sooner if symptoms worsen or unable to tolerate medication.",
    }
  ]
}
```

```

    "recorded_at": "2026-02-27T03:56:57.000+07:00",
    "recorded_by": "Dr.Sochea",
    "title": "Doctor"
  }
]
}

```

6.11 PRESCRIPTIONS

A prescription contains all medications, dosages, and instructions for a patient during a particular encounter.

6.11.1 Data Model

Field	Type	Require	Description
prescriptions	array of object	NO	Array of medications ordered by a doctor for a patient (nullable)

6.11.2 Prescription Model

Field	Type	Require	Description
patient_code	string	YES	Reference to the patient's unique code
visit_code	string	YES	Reference to visit the unique code
encounter_code	string		Reference to encounter a unique code
code	string	YES	Unique identifier for the prescription
prescribed_at	datetime	YES	Date and time the prescription was written
prescribed_by	string	YES	Name of healthcare provider who wrote the prescription
title	string		Position, whether doctor or nurse, or else
medications	array of objects	YES	Array of medications is prescribed

NOTE:

- ✎ **Medication dispenses during hospitalization:** Require encounter_code linking to progress_notes, outpatient, inpatient, surgery, or emergency
- ✎ **Medication dispensed for home use:** Require encounter_code to be null

6.11.3 Medication Model

Field	Type	Require	Description
medicine_code	string	YES	Unique identifier for the medication
medicine_name	string	YES	Name of the medicine
strength	string		Strength of the medicine

method	string	YES	Method of administration
unit	string	YES	Unit of measurement
morning	numeric		Morning dose
afternoon	numeric		Afternoon dose
evening	numeric		Evening dose
night	numeric		Night dose
days	integer	YES	Duration of prescription in days
interval	string		Interval between doses
note	text		Additional notes or instructions

NOTE:

- ✎ If no **interval** => **morning, afternoon, evening, night** should be presented
- ✎ Strength is required if it is not presented in medicine_name

6.11.4 Dispense Model

Field	Type	Require	Description
unit	string	YES	Unit of measurement (e.g., "Boxes", "Bottles", "Tablets")
quantity	integer	YES	Total amount dispensed to the patient

NOTE:

- ✎ Dispense: Specifies the total quantity of the drug provided to the patient.
- ✎ If the dispense field is not provided, the system calculates the quantity using this fallback logic:
 1. **Quantity Formula:** (morning + afternoon + evening + night) × days
 2. **Zero Fallback:** If days is 0 or missing, the quantity defaults to 0
 3. **Unit:** Defaults to the unit specified in the usage instructions

Example Calculation:

1. Morning: 1, Afternoon: 0, Evening: 1, Night: 0
2. Days: 7
3. Calculated Quantity: $(1 + 0 + 1 + 0) \times 7 = 14$ tablets

6.11.5 Example

```
{
  "prescriptions": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "code": "RX20250317001",
      "prescribed_at": "2025-03-17T10:10:00.000+07:00",
      "prescribed_by": "Dr. Chhun Sovannarith",
    }
  ]
}
```

```

"title": "Doctor",
"medications": [
  {
    "medicine_code": "MED001",
    "medicine_name": "Artemether-Lumefantrine",
    "strength": "20/120mg",
    "form": "Tablet",
    "method": "Oral",
    "unit": "Tablet",
    "morning": 4,
    "afternoon": 0,
    "evening": 4,
    "night": 0,
    "days": 3,
    "interval": null,
    "note": "Take with food. Complete full course even if feeling better."
  },
  {
    "medicine_code": "MED002",
    "medicine_name": "Paracetamol",
    "strength": "500mg",
    "form": "Tablet",
    "method": "Oral",
    "unit": "Tablet",
    "morning": null,
    "afternoon": null,
    "evening": null,
    "night": null,
    "days": 3,
    "interval": "8 hours",
    "note": "Take as needed for fever or pain.",
    "dispense": {
      "quantity": 1,
      "unit": "Boxes"
    }
  }
]
}

```

6.12 REFERRAL

Referral contains information about patient transfers between healthcare facilities.

6.12.1 Data Model

Field	Type	Description
referred_from	object	Information about the referring facility
referred_to	object	Information about the receiving facility

6.12.2 Detail Model

Field	Type	Require	Description
code	string	YES	Unique identifier for the referral
referral_number	string	YES	Unique reference number for the referral
transportation	string	YES	Mode of transportation
reason	string	YES	Reason for referral
has_called	boolean		Whether the receiving facility was called in advance
caretaker_name	string		Name of the person accompanying the patient
caretaker_phone	string		Contact number for the caretaker
referred_by	string		Name of the referring provider
referred_by_phone	string		Contact number for the referring provider
referred_at	datetime		Date and time of referral
received_by	string		Name of the receiving provider
received_by_phone	string		Contact number for the receiving provider
received_at	datetime		Date and time the patient was received
medications	string		Medications that the patient is taking
health_facility	object		receiving facility/ referring facility

6.12.3 Health Facility Model

Field	Type	Require	Description
name	string	YES	Health facility 's name
code	string	YES	Health facility 's code
label	string		Health facility 's label
type	string		Health facility 's type

6.12.4 Example

```
{
  "referrals": {
    "referred_from": null,
    "referred_to": {
      "code": "REF20250317001",
      "referral_number": "HCPP/REF/2025/317",
      "transportation": "Ambulance",
      "health_facility": {
        "name": "string", //required
        "code": "string", //required
        "label": "string", //optional
        "type": "HF level" //optional
      },
      "reason": "Severe malaria with signs of organ dysfunction
requiring ICU care",
    }
  }
}
```

```

    "has_called": true,
    "caretaker_name": "Sok Channary",
    "caretaker_phone": "012345678",
    "referred_by": "Dr. Chhun Sovannarith",
    "referred_by_phone": "0987654321",
    "referred_at": "2025-03-17T11:30:00.000+07:00",
    "received_by": "Dr. Pich Vothy",
    "received_by_phone": "0712345678",
    "received_at": "2025-03-17T12:15:00.000+07:00",
    "medications": "Artemether-Lumefantrine 20/120mg (1 dose given at
10:15), IV Artesunate 120mg initiated at 11:00"
  }
}
}

```

6.13 CONSULTATIONS

A clinical encounter where a patient interacts with a healthcare provider for assessment, advice, or diagnosis, *without necessarily receiving a prescription or procedure*.

Field	Type	Require	Description
consultations	Array of String	NO	assesses, advises, and guides a patient regarding their health condition.

Example:

```

{
  "visits": [
    {
      "health_facility_code": "HEFA-00002S75",
      "patient_code": "PT005852",
      "code": "VIS012455",
      "admission_type": "Walk-in",
      "visit_type": "OPD",
      "admitted_at": "2025-11-28T21:46:00.000+07:00",
      "discharge_type": "Authorizes",
      "discharged_at": "2025-11-28T22:23:00.000+07:00",
      "consultations": ["Do exercise regularly", "Drink more water"],
      {...}
    }
  ]
}

```

6.14 INVOICES

Invoice records track payments for healthcare services and medications.

6.14.1 Data Model

Field	Type	Require	Description
patient_code	string	YES	Reference to patient unique code
visit_code	string	YES	Reference to visit unique code
encounter_code	string		Reference to encounter unique code
code	string	YES	Invoice identifier
payment_type	string	YES	Type of payment
invoice_date	datetime	YES	Invoice date
total	number	YES	Total amount
created_at	datetime	YES	Creation timestamp
updated_at	datetime		Update timestamp
cashier	string	YES	Name of cashier

6.14.2 Service Model

Field	Type	Require	Description
invoice_code	string	YES	Invoice identifier
service_code	string	YES	Code identify the provided service
service_name	string	YES	Name of the service provided
service_category	string	YES	Category name of the service
price	number	YES	Unit price
quantity	number		Number of service provided
payment	number		Payment received from patient
paid	boolean		Payment status
discount_type	string		String whether in percentage or amount
discount	number		Discount amount or percentage

6.14.3 Medication Model

Field	Type	Require	Description
invoice_code	string	YES	Invoice identifier
medicine_code	string	YES	Code identifies the provided service
medicine_name	string	YES	Name of the service provided
quantity	number	YES	Number of medicines provided
price	number	YES	Unit price
payment	number		Payment received from patient
paid	boolean		Payment status

discount_type	string		String whether in percentage or amount
discount	number		Discount amount or percentage

6.14.4 Example

```
{
  "invoices": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "code": "INV20250317001",
      "payment_type": "HEF",
      "invoice_date": "2025-03-17T11:00:00.000+07:00",
      "total": 120000,
      "created_at": "2025-03-17T11:00:00.000+07:00",
      "updated_at": "2025-03-17T11:00:00.000+07:00",
      "cashier": "Sok Kunthea",
      "services": [
        {
          "invoice_code": "INV20250317001",
          "service_code": "SRV001",
          "service_name": "Blood Smear for Malaria",
          "service_category": "Lab Test",
          "quantity": 1,
          "price": 15000,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        }
      ],
      "medications": [
        {
          "invoice_code": "INV20250317001",
          "medicine_code": "MED001",
          "medicine_name": "Artemether-Lumefantrine 20/120mg",
          "quantity": 24,
          "unit": "Tablet",
          "price": 1500,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        },
        {
          "invoice_code": "INV20250317001",
          "medicine_code": "MED002",
          "medicine_name": "Paracetamol 500mg",
          "quantity": 24,
          "unit": "Tablet",
          "price": 1000,
          "payment": 0,

```

```

        "paid": false,
        "discount_type": null,
        "discount": 0
    }
]
}
]
}

```

7 SPECIFIC ENCOUNTERS

7.1 INPATIENT ADMISSION

Inpatient Admission represents a patient's hospital stay where at least one overnight stay is required for treatment, observation, or procedures.

7.1.1 Data Model

Field	Type	Require	Description
visit_code	string	YES	Reference to the visit unique code
is_new	boolean		Field to distinguish new cases from follow-ups
code	string	YES	Unique identifier for the operation
symptoms	Array of string		Primary symptom or problem reported by the doctor(nullable)
ward	string	YES	Name of the ward or clinic, or sub-clinic
service_type	string	YES	Name of the service or purpose of inpatient (nullable)
started_at	datetime	YES	Date and time operation started
ended_at	datetime		Date and time the operation ended
encountered_by	string	YES	Name of the healthcare provider
title	string		Position, whether doctor or nurse, or else
created_at	datetime	YES	Record creation timestamp
updated_at	datetime	YES	Record the last update timestamp

7.1.2 Example

```

{
  "inpatients": [
    {
      "visit_code": "V420650",
      "is_new": false,
      "code": "E317001",

```

```

    "symptoms": ["fever", "red skin"],
    "ward": "General IPD",
    "service_type": "Internal Medicine",
    "started_at": "2025-03-20T14:00:00.000+07:00",
    "ended_at": "2025-03-23T10:00:00.000+07:00",
    "encountered_by": "Dr. Chhun Sovannarith",
    "title": "Doctor",
    "created_at": "2025-03-20T14:00:00.000+07:00",
    "updated_at": "2025-03-23T10:00:00.000+07:00"
  }
]
}

```

7.2 OUTPATIENT

Outpatient represents a patient visit for treatment or consultation without requiring an overnight hospital stay.

7.2.1 Data Model

Field	Type	Require	Description
visit_code	string	YES	Reference to the visit unique code
is_new	boolean		Field to distinguish new cases from follow-ups
code	string	YES	Unique identifier for the operation
symptoms	Array of string		Primary symptom or problem reported by the doctor(nullable)
ward	string	YES	Name of the ward or clinic, or sub-clinic
service_type	string	YES	Name of the service or purpose of outpatient (nullable)
started_at	datetime	YES	Date and time the operation started
ended_at	datetime		Date and time the operation ended
encountered_by	string	YES	Name of the healthcare provider
title	string		Position, whether doctor or nurse, or else
created_at	datetime	YES	Record creation timestamp
updated_at	datetime	YES	Record the last update timestamp

7.2.2 Example

```

{
  "outpatients": [
    {

```

```

    "visit_code": "V420650",
    "code": "E317000",
    "symptoms": ["fever", "red skin"],
    "ward": "General Inpatient",
    "service_type": "General Medicine",
    "started_at": "2025-03-17T09:35:00.000+07:00",
    "ended_at": "2025-03-17T10:15:00.000+07:00",
    "encountered_by": "Dr. Chhun Sovannarith",
    "title": "Doctor",
    "created_at": "2025-03-20T14:00:00.000+07:00",
    "updated_at": "2025-03-23T10:00:00.000+07:00",
    "created_at": "2025-03-17T09:35:00.000+07:00",
    "updated_at": "2025-03-17T10:15:00.000+07:00"
  }
]
}

```

7.3 EMERGENCY

Emergency represents a patient visit requiring immediate attention for acute or potentially life-threatening conditions.

7.3.1 Data Model

Field	Type	Required	Description
visit_code	string	YES	Reference to the visit unique code
is_new	boolean		Field to distinguish new cases from follow-ups
code	string	YES	Unique identifier for the operation
symptoms	Array of string		Primary symptom or problem reported by the doctor(nullable)
ward	string	YES	Name of the ward or clinic, or sub-clinic
service_type	string	YES	Name of the service type or purpose (nullable)
started_at	datetime	YES	Date and time the encounter started
ended_at	datetime		Date and time the encounter ended
emergency_note	text	YES	Detailed notes about the emergency (nullable)
encountered_by	string	YES	Name of the healthcare provider
title	string		Position, whether doctor or nurse, or else
created_at	datetime	YES	Record creation timestamp
updated_at	datetime	YESs	Record the last update timestamp

7.3.2 Example

```
{
  "emergencies": [
    {
      "visit_code": "V420650",
      "code": "E317001",
      "is_new": false,
      "ward": "Emergency",
      "symptoms": ["None"],
      "service_type_name": "Emergency Medicine",
      "started_at": "2025-03-17T15:30:00.000+07:00",
      "ended_at": "2025-03-17T17:45:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00",
      "emergency_notes": "Patient brought by family after motorcycle accident. GCS 14/15. No loss of consciousness. Skull X-ray negative for fracture. CT not available. Multiple abrasions on right side. No apparent fractures. Wound cleaning and dressing performed. Tetanus prophylaxis given.",
      "created_at": "2025-03-17T15:30:00.000+07:00",
      "updated_at": "2025-03-17T17:45:00.000+07:00"
    }
  ]
}
```

7.4 SURGERY

Surgery refers to medical procedures performed on a patient in an operating theatre to treat, repair, or diagnose a condition, usually under anesthesia.

7.4.1 Data Model

Field	Type	Required	Description
visit_code	string	YES	Reference to the visit unique code
is_new	boolean		Field to distinguish new cases from follow-ups
encounter_code	string		Unique identifier for the encounter result in operation
code	string	YES	Unique identifier for the operation
symptoms	Array of string		Primary symptom or problem reported by the doctor(nullable)
theater_name	string	YES	Name of the operating theater
service_type	string	YES	Name of the operation
started_at	datetime	YES	Date and time the operation started

ended_at	datetime	YES	Date and time the operation ended
reason	string	YES	Reason for operating
anesthesia_type	string		Type of anesthesia used
procedure_notes	text	YES	Detailed notes about the procedure
complications	array of strings		Any complications during the operation
specimens	array of strings		Specimens collected during the operation
pre_operative_diagnosis	string		The clinical diagnosis made BEFORE surgery, based on symptoms, physical examination, and investigations.
post_operative_diagnosis	string		The final diagnosis confirmed AFTER the surgery, based on what the surgeon actually found during the operation.
blood_loss	string		Amount of blood loss during surgery (including unit)
surgeon_name	string	YES	Name of the surgeon
anesthetist_name	string		Name of the anesthetist
assistant_names	array of strings		Names of assistants
created_at	datetime	YES	Record creation timestamp
updated_at	datetime		Record the last update timestamp
details	object		Details how the surgery was performed

7.4.2 Detail Model

Field	Type	Required	Description
position	string	YES	How the patient lies during surgeries

7.4.3 Example

```
{
  "surgeries": [{
    "visit_code": "V420650",
    "encounter_code": "E401001",
    "code": "SUR20250401001",
    "theater_name": "Operating Theater 2",
    "service_type": "Laparoscopic Cholecystectomy",
    "started_at": "2025-04-01T08:30:00.000+07:00",
    "ended_at": "2025-04-01T10:15:00.000+07:00",
    "reason": "Symptomatic gallstones with recurrent biliary colic",
    "anesthesia_type": "General Endotracheal",
  }]
```

```

    "procedure_notes": "Patient positioned supine. Pneumoperitoneum
established with Veress needle. Four ports were placed. Gallbladder
dissected from liver bed with hook cautery. Cystic duct and artery
identified, clipped and divided. Gallbladder removed through umbilical
port. Hemostasis confirmed. No bile leak. Ports removed under direct
visualization. Wounds closed with 3-0 Vicryl and 4-0 Monocryl. Patient
tolerated procedure well.",
    "complications": [
        "Mild bleeding from liver bed, controlled with electrocautery"
    ],
    "specimens": [
        "Gallbladder with multiple stones",
        "Cystic fluid sample"
    ],
    "pre_operative_diagnosis": "Ancienne Sign nail du fémur gauche avec
plaie infecté (3ans)",
    "post_operative_diagnosis": "Ancienne Sign nail du fémur gauche avec
plaie infecté (3ans)",
    "details": {
        "position": "Décubitus dorsale"
    },
    "blood_loss": "5cl",
    "surgeon_name": "Dr. Sok Vichet",
    "anesthetist_name": "Dr. Meas Sokunthea",
    "assistant_names": ["Dr. Chhan Rathana" , "Nurse Van Bopha"s],
    "created_at": "2025-04-01T08:30:00.000+07:00",
    "updated_at": "2025-04-01T10:20:00.000+07:00"
  }
}

```

7.5 PROGRESS NOTES

Progress Note documents a patient’s ongoing assessment, treatment, and response during their care. It often uses the SOAP format.

7.5.1 Data Model

Field	Type	Require	Description
visit_code	string	YES	Reference to the visit unique code
encounter_code	string		Reference to the parent encounter code
code	string	YES	Unique identifier for the daily progress note
started_at	datetime	YES	Date and time the assessment started
ended_at	datetime		Date and time the assessment ended
encountered_by	string	YES	Name of the healthcare provider
title	string	YES	Position, whether doctor or nurse, or else
created_at	datetime	YES	Record creation timestamp
updated_at	datetime	YES	Record the last update timestamp

7.5.2 Example

```
{
  "progress_notes ": [
    {
      "visit_code": "VS20250317001",
      "encounter_code": "E417001",
      "code": "E317001",
      "started_at": "2025-03-17T16:00:00.000+07:00",
      "ended_at": "2025-03-17T16:15:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title ": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00"
    }
  ]
}
```

Subject to be changed by NPCA

8 SPECIFIC CLINICAL DATA FORM COMPONENTS

8.1 INTRODUCTION TO CLINICAL DATA FORM

Clinical Data Forms are designed to provide a standardized structure for collecting additional clinical information that cannot be adequately represented by the common clinical data components such as Patient, Visit, Triage, Vital Signs, Medical History, Diagnosis, Prescription, Referral, or Invoice.

A unique Form Code identifies each Clinical Data Form and contains a defined set of clinical attributes. The Form Code identifies the type of clinical form and determines how the submitted data should be processed and stored.

8.1.1 Form Structure

A Clinical Data Form must contain the following information:

Field	Type	Require	Description
visit_code	String	YES	Reference to the visit unique code
encounter_code	String	YES	A unique encounter identifier that references the code of the related clinical encounter. Its value must match exactly one of the following, depending on the encounter type: outpatient.code or inpatient.code
form_code	String	YES	Code of the clinical form.

form_name	String	YES	Name of the clinical form.
form_data	Object	YES	Clinical data collected by the form.

8.1.2 Example

```
"forms": [
  {
    "visit_code": "VIS071315",
    "form_code": "ENTY-000016Z",
    "form_name": "OPD Contraception",
    "encounter_code": "MED062210",
    "form_data": {
      "birth_control_type": "ថ្នាំគ្រាប់",
      "birth_control_customer_type": "ធ្លាប់ប្រើ"
    }
  }
]
```

8.2 DELIVERY FORM

The Delivery Form is used to record clinical information related to the delivery of a baby, including the delivery date and time, mode of delivery, maternal condition, delivery complications, and newborn information.

The form should be submitted when a delivery occurs or when the delivery event is clinically documented in the EMR.

8.2.1 Data Form Identifications

Name	Type	Require	Unit	Description
maternal_delivery_status	String			ស្ថានភាពស្ត្រីសម្រាលកូន
delivery_number_of_newborn	Number	YES		ចំនួនទារកទើបនឹងកើត
delivery_number_of_pregnant	Number	YES		ចំនួនមានផ្ទៃពោះ
delivery_number_of_delivery	Number	YES		ចំនួនសម្រាលកូន
delivery_number_of_miscarriage	Number	YES		ចំនួនរលូតកូន
delivery_number_of_abortion	Number	YES		ចំនួនរំលូតកូន
delivery_pregnancy_in_weeks	Number	YES	សប្តាហ៍	អាយុផ្ទៃពោះគិតជាសប្តាហ៍
delivery_at	Datetime	YES		កាលបរិច្ឆេទសម្រាល
delivery_type	String	YES		ប្រភេទសម្រាលកូន
delivery_oxytocin_injection	Boolean			ចាក់ Oxytocin ដល់ស្ត្រី (ក្នុងដំណាក់កាលទី៣នៃការសម្រាល)
delivery_oxytocin_injection_date	Datetime			កាលបរិច្ឆេទចាក់ Oxytocin
delivery_baby_born	String	YES		ទារកកើត

delivery_baby_breathing_in_first_minute	String			ទារកដកដង្ហើមក្នុង១នាទីដំបូង
delivery_baby_sex	String	YES		ទារកភេទ
delivery_breastfeed_in_first_hour	Boolean			បំបៅដោះកូនក្នុង១ម៉ោងដំបូង
delivery_baby_weight	Number	YES	g	ទម្ងន់ទារក (គិតជា g)

8.2.2 Validation Rule

Rule	Validation
Number of newborns	Must be ≥ 1
Delivery Datetime	Must not be later than the current date
Delivery Type	Must use valid reference data
Baby Born	Must use valid reference data.
Baby Sex	Must use valid reference data.

8.3 OPD ANC FORM

The OPD ANC Form is used to record clinical information related to antenatal care provided to a pregnant woman during an outpatient visit.

The OPD ANC Form should be submitted when an outpatient encounter includes antenatal care (ANC).

8.3.1 Data Form Identifications

Name	Type	Require	Unit	Description
delivery_number_of_pregnant	Number	YES		ចំនួនមានផ្ទៃពោះ
pregnancy_age	Number	YES	សប្តាហ៍	អាយុផ្ទៃពោះ (គិតជាសប្តាហ៍)
uterine_height_cm	Number		cm	កម្ពស់ស្បូន (ស.ម.)
fetal_heart_rate	Number		bpm	ចង្វាក់បេះដូងកូន
pregnancy_last_menstrual_period_date	Date			ថ្ងៃមករដូវចុងក្រោយ

8.3.2 Validation Rule

Rule	Validation
Number of Pregnancies	Must be ≥ 1
Pregnancy Age	Integer/decimal, clinically valid range
Uterine Height	Numeric, ≥ 0 ; recommended to define a reasonable maximum
Fetal Heart Rate	Required when fetal heart rate is assessed; numeric; range: 60–220 bpm
Last Menstrual Period Date	Required if known; must be a valid date; must not be in the future ; should be before or equal to the visit date

8.4 OPD PNC FORM

The OPD PNC (Postnatal Care) Form records postnatal visit information for both the mother and the newborn after delivery.

8.4.1 Data Form Identifications

Name	Type	Require	Description
pnc_days_after_delivery	Number	YES	ចំនួនថ្ងៃក្រោយសម្រាលកូន
delivery_at	Datetime	YES	កាលបរិច្ឆេទសម្រាលកូន
pnc_maternal_danger_signs_and_treatment	String	YES	សញ្ញាគ្រោះថ្នាក់របស់ម្តាយ
pnc_baby_danger_signs_and_treatment	String	YES	សញ្ញាគ្រោះថ្នាក់របស់ទារក

8.4.2 Validation Rule

Rule	Validation
Days After Delivery	The value must be calculated from the delivery date and visit date.
Delivery Datetime	Must not be later than the visit date

8.5 ABORTION FORM

The Abortion Clinical Data Form records the patient's pregnancy/obstetric history, assessment at arrival, contraceptive service, and the abortion method provided during the encounter.

8.5.1 Data Form Identifications

Name	Type	Require	Unit	Description
children_alive	Number	YES		ចំនួនកូនរស់
delivery_number_of_misconriage	Number	YES		ចំនួនលូតកូន
delivery_number_of_abortion	Number	YES		ចំនួនលូតកូន
pregnancy_pregnancy_test	String			ការធ្វើតេស្តមានផ្ទៃពោះ
problems_upon_arrival	Boolean			អតិថិជនមានបញ្ហាទេ ពេលមកដល់
type_of_contraceptive_method_received	String			តើអតិថិជនបានទទួលមធ្យោបាយពន្យារកំណើតប្រភេទណា ?
safe_abortion_is_customer_get_method_before_leave	String	YES		តើអតិថិជនបានទទួលមធ្យោបាយពន្យារកំណើតមុនចេញទៅផ្ទះដែលឬ ទេ ?
safe_abortion_uterine_size	Number	YES	សប្តាហ៍	ទំហំស្បូនគិតជាសប្តាហ៍
safe_abortion_method	String	YES		វិធីសាស្ត្រលូតកូន

8.5.2 Validation Rule

Rule	Validation
Children Alive	Must be >= 0
Number of Miscarriages	Must be >= 0
Number of Abortions	Must be >= 0
Pregnancy Test	Must use valid reference data
Type of Contraceptive Method Received	Must use valid reference data
Did the client receive birth control before going home?	Must use valid reference data
Uterine size	Must be >= 0, must be a valid numeric value
Abortion method	Must use valid reference data

8.6 BIRTH CONTROL FORM

The Birth Control Form is a clinical data form used to record family planning and contraceptive services provided to a client. It captures the client's selected contraceptive method, counseling, provision of the method, and relevant follow-up information.

The form can be used during a family-planning visit or as part of postpartum/post-abortion care when contraception is provided.

8.6.1 Data Form Identifications

Name	Type	Require	Description
birth_control_type	String	YES	ប្រភេទពន្យារកំណើត
birth_control_customer_type	String	YES	ប្រភេទពន្យារកំណើតរបស់អតិថិជន

8.6.2 Validation Rule

Rule	Validation
birth_control_type	Must use valid reference data
birth_control_customer_type	Must use valid reference data

8.7 CHILD PHYSICAL EXAMINATION (0-28 DAYS) FORM

The Child Physical Examination (0–28 Days) form is used to document the physical examination of a newborn from birth up to 28 days of age. It focuses on detecting congenital abnormalities, birth complications, infection, feeding problems, and other neonatal danger signs.

8.7.1 Data Form Identifications

Name	Type	Require	Unit	Description
pnc_days_after_delivery	Number	YES		ចំនួនថ្ងៃក្រោយសម្រាលកូន

child_growth_born_weight	Number	YES	g	ទម្ងន់ពេលកើត (គិតជា g)
baby_maternal_fever_at_birth	String			ម្តាយក្តៅខ្លួនខ្លាំងនៅពេលសម្រាល
baby_long_labor	String			ការសម្រាលមានរយៈពេលយូរ
term_or_preterm_birth	Boolean	YES		កើតគ្រប់ខែ/មិនគ្រប់ខែ
baby_apgar_one_minute	Number		នាទី	APGAR (1 នាទី)
baby_apgar_ten_minutes	Number		នាទី	APGAR (10 នាទី)
baby_apgar_five_minutes	Number		នាទី	APGAR (5 នាទី)
baby_born_with_clubfoot	String			ទារកកើតបញ្ឆោតជើង
difficulty_with_breastfeeding	String			បញ្ហាក្នុងការបៅ
baby_seizures	String			ប្រកាច់ (ទារកប្រកាច់)
meconium_passed_within24Hours	String			គ្មានអាចម៍ព្រៃក្នុងកំឡុងពេល២៤ម៉ោងដំបូង
examination_results	String	YES		លទ្ធផលនៃការពិនិត្យ
other_observations	String			ការសង្កេតផ្សេងៗ(ការកម្សាយផ្សេងៗ)
transfer	String			បញ្ជូន

8.7.2 Validation Rule

Rule	Validation
Baby Maternal Fever at birth	Must use valid reference data
Baby long labor	Must use valid reference data
Baby Premature birth	Must use valid reference data
Baby Seizures	Must use valid reference data
Baby Born with Clubfoot	Must use valid reference data

8.8 CHILD PHYSICAL EXAMINATION (1MONTH-5YEARS) FORM

Child Physical Examination (1 Month–5 Years) is a clinical form used by healthcare providers to document the physical examination findings of children from 1 month up to 5 years of age.

The purpose is to identify normal or abnormal physical findings, monitor the child's health and development, and support diagnosis and treatment.

8.8.1 Data Form Identifications

Name	Type	Require	Unit	Description
child_had_serious_health_problems	String	YES		កុមារធ្លាប់មានបញ្ហាសុខភាពធ្ងន់ធ្ងរ

child_growth_born_weight	Number	YES	g	ទម្ងន់ពេលកើត (គិតជា g)
term_or_preterm_birth	Boolean	YES		កើតគ្រប់ខែ/មិនគ្រប់ខែ
complication_during_childbirth	Boolean			មានផលវិបាកពេលសម្រាល
feeding_difficulties	Boolean			បញ្ហាក្នុងការហូប/ផឹក
child_convulsions	Boolean			ប្រកាច់
toilet_training	String			ការចេះបន្ទោរបង់(ធៀបនឹងកុមារផ្សេងដែលមានអាយុដូចគ្នា)
difficulty_with_breastfeeding	Boolean			បញ្ហាក្នុងការបៅ
child_speech_development	String			ការនិយាយ
social_play_development	String			ការលេង(ធៀបនឹងកុមារផ្សេងដែលមានអាយុដូចគ្នា)
examination_results	String	YES		លទ្ធផលនៃការពិនិត្យ
other_observations	String			ការសង្កេតផ្សេងៗ(ការកម្រោយផ្សេងៗ)
transfer	String			បញ្ជូន

8.8.2 Validation Rule

Rule	Validation
Child Convulsions	Must use valid reference data

9 STANDARD DATA REFERENCE GLOSSARY

9.1 PATIENT DEMOGRAPHIC

This section specifies the standardized patient information that should be recorded.

9.1.1 Sex

Code	Value	Description
F	F	Patient who is a female from birth
M	M	Patient who is a male from birth

9.1.2 Marital Status

Code	Value	Description
MAST-00M002	Married	Patient is legally married
MAST-0000X1	Single	Patient has never been married
MAST-000S03	Widowed	Patient's spouse is deceased
MAST-0000N4	Divorced	Patient has been legally divorced and has not remarried
MAST-0000N5	Separated	Patient is legally married but living apart from their spouse

9.2 VITAL SIGNS OBSERVATIONS

This section defines the standard vital signs recorded during patient assessment.

Concept	Description
Temperature	A float value representing the body's core thermal state.
Heart Rate	An integer that functions as a real-time counter for the heart's pump cycle, measured in beats per minute (bpm).
Respiratory Rate	An integer tracking the frequency of the body's primary resource-intake cycle (breathing).
Systolic Blood Pressure	An integer that measures the peak pressure on the circulatory network during a pump cycle.
Diastolic Blood Pressure	An integer that measures the baseline pressure when the pump is in its resting phase.
Blood Oxygen	A percentage integer that measures the saturation of the body's data carriers (hemoglobin) with oxygen.
Random Blood Glucose	An integer representing the amount of readily available fuel (glucose) in the system.

9.3 VISIT & ADMINISTRATIVE DATA

This section covers categorical data related to patient visits, admissions, and billing.

9.3.1 Admission

Code	Value	Description
ADTY-000J008	Walk-in	The patient comes to the facility without a prior appointment.
ADTY-000009S	Appointment	Patient arrives for a scheduled visit.
ADTY-0P0000A	Refer	Patient is referred by another facility or provider.

9.3.2 Visit Types

Code	Value	Description
VITY-00J000C	IPD	Patient stays at the hospital for one or more nights.
VITY-00L000D	OPD	Patient receives care without being admitted overnight.

9.3.3 Outcome Types

Value	Description
Unchanged	No significant change in the patient's condition.
Recovered	Patient's condition has resolved.
Improved	The condition has shown improvement but is not fully resolved.
Hopeless	The patient's condition has no viable solution.
Home treatment	The patient can continue to receive treatment at home.
Deceased	The patient passed away during the visit.

9.3.4 Discharge Types

Value	Description
Authorized	Patient is well enough to go home.
Unauthorized	Patient chose to leave before formal discharge.
Deceased	Patient passed away during the stay.
Refer Out	Patient is sent to another hospital or care facility.
Absconded	Patient left without informing staff or completing a formal discharge.

9.3.5 Diagnosis Types

Value	Description
Differential	A list of possible conditions that could explain the symptoms.
In	The condition is suspected when the patient is first admitted.
Out	The final diagnosis after treatment and evaluation was recorded at discharge.
Primary	The main reason for the current hospital visit or treatment.
Secondary	Coexisting conditions that influence care or outcomes.

9.3.6 Payment Types

Code	Value	Description
PATY-0000N09	HEF (Poor)	The patient is from a poor family, and payment is covered by the HEF
PATY-00002NF	HEF (Informal Worker)	Patient is an informal worker with payment covered by the HEF
PATY-00003LD	HEF (At Risk)	Patient is from a family at risk, and payment is covered by the HEF
PATY-0000G3E	NSSF Employee	Patient is an employee with payment is covered by the NSSF
PATY-00B003F	NSSF Public Servant	Patient is a public servant with payment is covered by the NSSF
PATY-000T040	NSSF Self-Employed	Patient is self-employed, with payment is covered by the NSSF

9.3.7 Card Types

Value	Description
HEF	Health Equity Fund for Poor family
HEFR	Health Equity Fund for At Risk family
HEFIY	Health Equity Fund for Informal Worker with Yellow Card
HEFIV	Health Equity Fund for Informal Worker with Violet Card
NSSF	National Social Security Fund
NSSP	National Social Security Fund for Public servant
NSSPM	National Social Security Fund for Public servant's family Member
NSSE	National Social Security Fund for Employees
NSSEM	National social Security Fund for Employees' family Member
NSSFS	National Social Security Fund for Self-employed person
NSSFSM	National Social Security Fund for Self-employed person's family Member

9.4 MEDICAL HISTORY COMPONENTS

This section outlines the key areas to be documented when taking a patient's medical history.

- **History of Present Illness:** A detailed narrative of symptoms—onset, duration, severity, aggravating/relieving factors, and associated symptoms.
- **Past Medical History:** Previous diagnoses (e.g., diabetes, hypertension), surgeries, hospitalizations, and chronic conditions.
- **Past Surgical History:** Details of past operations, including type, date, and any complications.
- **Current Medication:** A list of current medications (name, dose, frequency), over-the-counter drugs, and supplements.
- **Allergies:** Known drug, food, or environmental allergies and the type of reaction.
- **Family History:** Hereditary conditions in parents, siblings, and grandparents (e.g., heart disease, cancer).
- **Social History:** Lifestyle factors such as tobacco use, alcohol, drug use, occupation, living situation, and marital status.
- **Obstetric & Gynecologic History:** For female patients: menarche, menstruation pattern, pregnancies, contraception, and menopause.
- **Immunization History:** Vaccination status, especially for tetanus, flu, COVID-19, and common childhood vaccines.

9.5 PHYSICAL EXAMINATION COMPONENTS

This section details the systems to be assessed during a physical examination and what to record for each.

- **General Appearance:** Consciousness (alert/drowsy), Distress, Nutrition, Posture, Gait, Hygiene, Pallor, Jaundice, Cyanosis, Edema.
- **HEENT:** Head, Eyes, Ears, Nose, Throat/Mouth.
- **Neck:** Lymph nodes, Thyroid gland, Tracheal position, Jugular Venous Pressure (JVP).
- **Cardiovascular System:** Inspection (chest), Palpation (apex beat), Auscultation (S1/S2, murmurs), Peripheral pulses.
- **Respiratory System:** Chest symmetry, Expansion, Percussion (dullness), Breath sounds, Crepitations, Wheezing.
- **Abdomen:** Inspection (distension), Palpation (tenderness, organ size), Percussion, Bowel sounds.
- **Musculoskeletal System:** Joint swelling, Deformity, Range of motion, Muscle tone/strength, Spine, Gait.
- **Nervous System:** Mental status, Cranial nerves, Motor/sensory function, Reflexes, Coordination.
- **Skin:** Rashes, Lesions, Color changes, Ulcers, Turgor, Temperature.
- **Lymph Nodes:** Cervical, Axillary, Inguinal – note any enlargement, tenderness, or changes in mobility.
- **Glasgow Coma Scale:** Structured consciousness assessment using Eye, Verbal and Motor responses, producing a GCS score from 3–15.
- **Urogenital System:** Genitourinary examination: urinary/genital findings, tenderness, swelling, discharge, lesions, etc.
- **Mental Status:** Cognitive and behavioral assessment: appearance, behavior, mood, speech, orientation, memory, attention, thought process, etc.
- **Other body parts:** Examination findings involving body parts or anatomical areas not covered by the standard examination sections above.

9.6 FORM DATA

9.6.1 Birth Control Type

Code	Value
OBVA-S000099	ថ្នាំគ្រាប់ស៊ីអូស៊ី
OBVA-00D0097	ថ្នាំចាក់
OBVA-00P0096	កងដាក់ក្នុងស្បូន
OBVA-00W009A	កងដាក់ក្រោមស្បែក
OBVA-0000C98	ស្រោមអនាម័យ
OBVA-000159Z	ដោះកងដាក់ក្នុងស្បូន
OBVA-00A015B	ការដាក់កងក្នុងដៃ
OBVA-000W15C	តាមដានកងដាក់ក្នុងដៃ
OBVA-C00015D	ដោះកងដៃចេញ
OBVA-00015VE	ការផ្តល់ប្រឹក្សា

9.6.2 Birth Control Customer Type

Code	Value
OBVA-00001H1	ធ្លាប់ប្រើ
OBVA-0000I10	លើកដំបូង
OBVA-0000I12	ចាស់

9.6.3 Delivery Baby Born

Code	Value
OBVA-00007ID	កើតរស់
OBVA-00T007C	កើតស្លាប់

9.6.4 Delivery Baby Sex

Code	Value
OBVA-00G007A	ប្រុស
OBVA-00007GB	ស្រី

9.6.5 Delivery Type

Code	Value
OBVA-000B153	សម្រាលធម្មតា

OBVA-00R0154	សម្រាលដោយអន្តរាគមន៍
OBVA-X000155	សម្រាលដោយវះកាត់

9.6.6 Safe Abortion Method

Code	Value
OBVA-00H0156	វិលូតដោយថ្នាំ
OBVA-000157X	វិលូតដោយបូមសុញ្ញាកាស
OBVA-0001Z58	វិលូតដោយពង្រីកមាត់ស្បូន និងយកចេញ
OBVA-000159Z	វិធីសាស្ត្រផ្សេងទៀត

9.6.7 Type Of Contraceptive Method Received

Code	Value
OBVA-S000099	ថ្នាំគ្រាប់ស៊ីអូស៊ី
OBVA-00D0097	ថ្នាំចាក់
OBVA-00P0096	កងដាក់ក្នុងស្បូន
OBVA-00W009A	កងដាក់ក្រោមស្បែក
OBVA-0000C98	ស្រោមអនាម័យ
OBVA-0000I9B	បញ្ឈប់កំណើត

10 RELATED DOCUMENTS

- EMR Clinical Data Components
- OPD Sample JSON
- IPD Sample JSON
- Transfer OPD to IPD Sample JSON
- Referral Sample JSON
- Form Sample JSON